# GrappStar base station build guide

04 November 2026

Subsystems: drum, frame, launch, reel, sensors. Model exported 2026-11-04 13:38.
All sizes in mm. Masses and print times are estimates.

## 1. Safety

- Wiring always starts with the battery unplugged. Plug it in last, after a check of every connection.
- Keep hands, hair and loose cloth away from the gears, the reel and the launcher while the motor can run.
- Never stand in front of the launcher. Treat it as loaded whenever the spring is cocked.
- Release the spring before you work on the launcher.
- Wear eye protection when you cut, drill or press parts, and when you test a launch.
- Let hot printed parts and the nozzle cool before you touch them. Print PPA-CF with the enclosure closed.
- Charge LiPo batteries on a fireproof surface and never leave them charging alone.

## 2. First print: test coupons

Print these before any big part. Each coupon uses the same geometry and numbers as the real part. Files: base/pack/coupons (3MF plates, STL, COUPON_TEST.pdf). Also print the fit coupon (python -m tools.coupon OUT) to tune cad/lib/fits.py.

Print these before the big parts. Each coupon is made from the same geometry functions and numbers as the real part, so a pass here is a pass there. Let parts cool 10 minutes before testing. The fit coupon (tolerance tiles A to H, base/pack/fit_coupon/COUPON.md) is printed alongside.

### Settings

PLA: 0.4 mm nozzle, 0.2 mm layers, 4 walls, 40 % gyroid infill unless noted, 5 top and 5 bottom layers, hole and XY compensation 0, no supports. PPA-CF: hardened steel nozzle (never brass), filament dried (80 C, 8 h) and fed from a dry box, 0.2 mm layers, 6 walls, 60 % gyroid infill, 6 top and 6 bottom layers, hole and XY compensation 0, no supports. The 3MF files carry geometry only: set the material and these settings in Bambu Studio.

### If a coupon fails

To change a number: on the Mac, after bash kedge/setup_mac.sh (from the repo root), cd base and run ../kedge/.venv/bin/python -m tools.risk_coupon (it rewrites base/pack/coupons and base/pack/fit_coupon). Easier: give the failing result to a Claude session and let it change the numbers. Change one number at a time and reprint only that coupon.

### Plates

plate material objects
risk_plate_1_PLA.3mf PLA 24
risk_plate_2_PPA-CF.3mf PPA-CF 26

### Tests

### 1. Swing-arm direction splitter

- Parts: rc1_plate, rc1_sleeve, rc1_sun, rc1_arm, rc1_drag_pad, rc1_drag_shim, rc1_idler_clip, rc1_idler, rc1_output, rc1_crank, rc1_stop, rc1_stop_e
- Bought: 4 x M3x16 (two stop fingers); everything else printed
- Do: Push the drag pad root block and the shim into the arm hub pocket (finger in its recess). On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, printed C clip into the pin groove, output on its pin, crank on top, the two stop fingers on their columns (2 x M3x16 each). Turn the crank 10 turns each way.
- Pass: Clockwise seen from above: the arm swings the idler into the output at once and comes to rest on the engaged stop finger, the output turns, no skipped teeth. Brake the output lightly with a finger and keep turning: the arm stays on that finger (the idler is held at full depth, never pulled in deeper) and the crank still turns smoothly, no binding. Anticlockwise: the arm swings out to the other finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement. The pad drags the arm reliably both ways and the clip stays on.
- If it fails, change: launch_aim.py: FIT (pin and bore play), DRAG_INT and DRAG_SHIM_ADJ (drag: or print a thicker shim), clips.py for the C clip, A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy (swung-out stop), _stop_post_e_xy (engaged stop: binding with the output braked means it lets the idler in too deep); launch_gears.py center_distance extra / spur backlash (mesh).

- Made from: launch_aim.sun_solid, arm_solid, idler_solid (m1.5 z30/18/30, face 12, FIT 0.35), launch_gears.spur for the output, launch_aim.A: centre distances 36.10 / 36.10, output at (11.7, 68.4), engaged arm 96.2 deg on the engaged stop (launch_aim._stop_post_e_xy), swung out 116.2 deg on the other stop (launch_aim._stop_post_xy), sleeve rib launch.RIB_W 5.8 / RIB_R 12.6; drag_pad_solid, drag_shim_solid (DRAG_INT 0.45, shim up to +0.25), pin_clip_solid

### 2. Printed contact switch

- Parts: rc2_switch_base, rc2_switch_lever

- Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter

- Do: Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.

- Pass: Measure the tip travel with calipers (depth rod from the lever tip down to the base top, note the reading at rest, press until it beeps, read again): beep within 0.8 to 1.4 mm of tip travel (design 1.14), silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.

- If it fails, change: cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.

- Made from: cad.lib.switch.contact_switch(lever_len=36) with the library defaults (travel 0.8, overtravel 0.6, beam t 1.6): gap 0.91, pretravel at the tip 1.14, close force 0.83 N, strain 0.0061

### 3. Dog clutch and compliant fork

- Parts: rc3_base, rc3_reel_face, rc3_collar, rc3_fork, rc3_dog_post, rc3_rail

- Bought: M3 nut (yoke tab), 2 x M3x12 (SW_DOG screws in the post), 1 x M3 (post clamp), multimeter

- Do: Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, dog post clamped on the rail below the fork, SW_DOG screws in the post set 0.5 mm short of the tab nut at full engagement. Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back. Meter on beep between the lower screw and the tab nut.

- Pass: Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat; the meter beeps only with the dogs fully in, never tip to tip.

- If it fails, change: _drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove, dog_touch, dogpost_s; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).

- Made from: drum.dog_teeth (dog_n 4, dog_h 5, dog_deg 36, backcut 4, roof 1.2, r 18..26), drum.fork, drum.dog_post (dog_touch 0.5), drum.rail, drum.hexprism, GEOM journal d33, groove (82.0, 88.0), stroke 7, neutral gap 2.5

### 4. Face slip clutch with the flexure spring

- Parts: rc4_hub_plate, rc4_friction_disc, rc4_pressure_plate, rc4_flexure, rc4_adjuster_plate

- Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, coins in a small bag, kitchen scale

- Do: Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches (zero), then turn all 3 nuts the same, 1/4 turn (0.125 mm) at a time. Clamp the stub level in a vice, tie a string to a disc lug (r 28.5 mm), lay it over the rim and hang a bag of coins weighed on a kitchen scale. Add or take coins to find the hold and slip weights at each setting.

- Pass: At each setting the lower weight must not pull the disc round and the higher one must (smooth steady slip, no judder); same result 3 times; the flexure never bottoms before 1 turn. Bands: 1/4 turn: 35 g holds, 60 g slips (design 48 g, 0.013 N m); 1/2 turn: 70 g holds, 120 g slips (design 95 g, 0.027 N m); 3/4 turn: 105 g holds, 180 g slips (design 143 g, 0.040 N m); 1 turn: 140 g holds, 240 g slips (design 190 g, 0.053 N m).

- If it fails, change: cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).

- Made from: cad.lib.clutch.slip_clutch() defaults (reel does not use it: its clutch is the spool face plus the PETG rocker): shaft d20 key 6, friction r 18.5..27, mu 0.3, flexure t 0.8 gap 0.5, 0.106 N m per mm, travel 0.50; on the shared keyed stub of coupon 7

### 5. Bearing seats

- Parts: rc5_seats_608_693, rc5_csk20_seat

- Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP

- Do: Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.

- Pass: Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.

- If it fails, change: fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. If the 608 is off, print tile A of the fit coupon (base/pack/fit_coupon, COUPON.md).

- Made from: cad.lib.bearings.seat_608(bottom lip, lip_d 14) as reel, seat_f693(), seat_693(), seat_csk20(key_deg 90) as the reel capstan block (wall 4); fits BRG_RIB_INTERF 0.15, BRG_RIB_BORE 0.3, CSK20_KEY_RCLEAR 0.2

### 6. M3 holes

- Parts: rc6_m3_tile

- Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10

- Do: Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.

- Pass: Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.

- If it fails, change: fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (tiles C, D, E of the fit coupon in base/pack/fit_coupon bracket them).

- Made from: cad.lib.fasteners.m3_insert (d 4.0), m3_nut_trap (AF 5.8), m3_clear (d 3.4), m3_tap (d 2.6)

### 7. Key rib and spigot

- Parts: rc7_key_stub, rc7_key_bore, rc7_spigot, rc7_bushing

- Bought: CSK20PP from coupon 5 (inner keyway)

- Do: Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.

- Pass: Stub slides in by hand, key in the keyway, no visible rock. Play: clamp the tile, tape a ruler to the stub end, mark the ruler 60 mm from the axis, rock the stub both ways: the mark moves under 1 mm (about 1 deg). The spigot turns freely; rocked at its free end it moves under 0.5 mm.

- If it fails, change: fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.

- Made from: cad.lib.shafts.keyed_shaft(d 20, key_w 6, key_h 2.6) and keyed_bore (clear 0.3); drum.main_shaft (spig_d 12); launch.sleeve_solid (SPIGOT_BORE 12.35)

### 8. Gear meshes m1.25, m1.5, m2, m2.5

- Parts: rc8_m125_plate, rc8_m125_seg, rc8_m125_pinion, rc8_m15_plate, rc8_m15_z42, rc8_m15_z24, rc8_m2_plate, rc8_m2_z50, rc8_m2_z22, rc8_m25_plate, rc8_m25_big, rc8_m25_drive

- Bought: none

- Do: Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.

- Pass: Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.

- If it fails, change: fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).

- Made from: m1.25: gears.spur_gear z60 sector / z10 shift 0.4 tip_short 0.08 (drum.tip_short_for), cd 44.33 (drum.a_seg); m1.5: gears.spur_gear z42 sector / z24, cd 49.60; m2: gears.spur_gear z50 sector (drum GEOM out_m, out_z) / z22 (reel Z_CAPG), cd 72.10; m2.5: launch_gears.sector_gear_2d z40 (n_present 27, last 6 + gap) / launch_gears.spur z26, cd 82.60 (launch L); coupon face 6, fits GEAR_BACKLASH 0.15, GEAR_CENTER_EXTRA 0.1

### 9. Shear-pin torque limiter

- Parts: rc9_hub_lever, rc9_outer_lever, rc9_shear_pin x3

- Bought: string or a strap, luggage scale

- Do: Print the pins lying down on their flat, as exported (D44: shear plane across the layers; never stand them up). Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.

- Pass: Each pin shears cleanly between 5.9 and 10.2 kg on the scale (3.5 to 6 N m, drum_NOTES; design 4.9 N m = 8.3 kg), the 3 pins within 20 percent of each other, both slices undamaged (holes still round); a fresh pin goes in by hand.

- If it fails, change: drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.

- Made from: drum.coupler_hub and drum.coupler (sliced s 4.5..20), drum.shear_pin: d4 PLA in single shear at r 13 (shear_s 9); 4.9 N m at 30 MPa

### 10. AS5600 magnet cap (bench test)

- Parts: rc10_magnet_cap

- Bought: magnet: not included with the board, see MEASURE_LIST C13 (any small neodymium disc edge-on, or 6 x 2.5 diametric), M3x6, superglue

- Do: Glue the magnet into the end pocket (flush, any way round), push the cap on the shaft end (flat to the flat), M3x6 onto the flat. It fits the d12 main shaft spigot of the base (for the bare-motor bench test, tape the magnet to the wiper shaft end instead). AS5600 chip on the shaft axis, 1 to 2 mm from the magnet face.

- Pass: Cap does not wobble or slip on the shaft; the AS5600 angle follows the shaft smoothly through full turns (no jumps), magnet status OK (not too weak, not too strong) at 1 to 2 mm.

- If it fails, change: drum.py GEOM cap_s, spig_d; magnet_holder pocket d6.2 x 2.6 (magnet loose or tight); fits.py HOLE_SLIDE (cap on the spigot).

- Made from: drum.magnet_holder (spig_d 12 with its flat, cap_s (195.0, 201.5), magnet pocket d6.2 x 2.6), printed as drum prints it

### Parts

part material qty print bbox mm what
rc1_plate PLA 1 110.1 x 82.2 x 33.1 Splitter base: centre pin d8, output pin d8, two stop finger columns (2 x M3 tap each)
rc1_sleeve PPA-CF 1 26.0 x 26.0 x 29.3 Sleeve stub d20 with the integral key rib (launch.RIB_W, RIB_R) under the sun and the crank
rc1_sun PPA-CF 1 47.8 x 48.0 x 12.0 launch_aim.sun_solid: sun m1.5 z30 as built
rc1_arm PPA-CF 1 61.4 x 41.0 x 19.3 launch_aim.arm_solid: swing arm as built (drag pad pocket, d8 pin, clip groove)
rc1_drag_pad PLA 1 24.6 x 8.7 x 4.0 launch_aim.drag_pad_solid: printed flexure drag pad as built
rc1_drag_shim PLA 1 9.4 x 3.0 x 4.0 launch_aim.drag_shim_solid: nominal preload shim as built
rc1_idler_clip PLA 1 10.8 x 10.0 x 1.5 launch_aim.pin_clip_solid: printed C clip for the d8 idler pin
rc1_idler PPA-CF 1 30.0 x 29.7 x 12.0 launch_aim.idler_solid: idler m1.5 z18 as built
rc1_output PPA-CF 1 47.8 x 48.0 x 12.0 Output m1.5 z30 (launch_gears.spur, face 12) on a d8 pin
rc1_crank PPA-CF 1 48.2 x 30.3 x 20.0 Hand crank, keyed on the sleeve rib like the sun
rc1_stop PLA 1 47.1 x 10.0 x 6.4 Stop finger: rounded end at the as-built swung-out stop post position
rc1_stop_e PLA 1 64.4 x 10.0 x 6.4 Engaged stop finger: rounded end at the as-built engaged stop post position (launch_aim._stop_post_e_xy)
rc2_switch_base PLA 1 54.0 x 35.0 x 11.5 contact_switch base: pad, nut pillar, hard stop, wire channel
rc2_switch_lever PLA 1 50.0 x 5.0 x 8.0 contact_switch lever, printed on its side (Y up)
rc3_base PLA 1 40.0 x 91.5 x 97.5 Base: d30 journal (REEL sleeve) and hex 22 AF (main shaft), rail hole (press)
rc3_reel_face PPA-CF 1 61.6 x 54.0 x 9.5 REEL gear dog face (drum.dog_teeth, phase 45) with a handle
rc3_collar PPA-CF 1 60.0 x 60.0 x 18.0 Collar lower half: drum.dog_teeth facing down, fork groove, hex bore

rc3_fork PLA 1 87.1 x 94.1 x 83.0 drum.fork as built: compliant yoke, rail sleeve, follower pin, M3 nut in the tab
rc3_dog_post PLA 1 20.0 x 75.9 x 41.7 drum.dog_post as built: SW_DOG_L / SW_DOG_R screw bosses, clamps on the rail
rc3_rail PPA-CF 1 110.0 x 10.0 x 9.0 drum.rail profile d10 with the print flat, cut to 110 long
rc4_hub_plate PLA 1 54.0 x 54.0 x 6.0 clutch.slip_clutch hub_plate (library defaults)
rc4_friction_disc PLA 1 57.0 x 55.0 x 3.0 clutch.slip_clutch friction_disc (library defaults)
rc4_pressure_plate PLA 1 54.0 x 54.0 x 3.5 clutch.slip_clutch pressure_plate (library defaults)
rc4_flexure PLA 1 50.0 x 50.0 x 1.7 clutch.slip_clutch flexure (library defaults)
rc4_adjuster_plate PLA 1 54.0 x 54.0 x 4.0 clutch.slip_clutch adjuster_plate (library defaults)
rc5_seats_608_693 PLA 1 60.0 x 28.0 x 9.6 608 (reel spool seat), F693ZZ and 693ZZ crush-rib seats
rc5_csk20_seat PLA 1 55.3 x 55.3 x 16.0 CSK20PP seat with the printed key (reel capstan block, key_deg 90)
rc6_m3_tile PLA 1 54.0 x 18.0 x 8.6 M3 heat-set hole, nut trap, clearance and tap holes (fits.py values)
rc7_key_stub PPA-CF 1 21.2 x 20.0 x 31.0 shafts.keyed_shaft d20 key 6 (reel capstan tube numbers), 31 long, also the coupon 4 clutch shaft
rc7_key_bore PPA-CF 1 36.0 x 36.0 x 8.6 shafts.keyed_bore d20 key 6, 8 long (keyed hubs on the sleeve rib are PPA-CF)
rc7_spigot PPA-CF 1 28.0 x 25.4 x 22.0 drum.main_shaft slice: hex 8 long + d12 spigot 20 long, on a hex flat
rc7_bushing PPA-CF 1 32.0 x 32.0 x 16.0 launch.sleeve_solid slice 16 long: spigot bore d12.35
rc8_m125_plate PPA-CF 1 62.3 x 18.0 x 11.0 Pins at 44.33
rc8_m125_seg PLA 1 38.1 x 46.7 x 6.0 drum segment gear m1.25 z60 sector
rc8_m125_pinion PPA-CF 1 15.8 x 15.2 x 6.0 drum wrap pinion m1.25 z10 x0.4 tip_short 0.08
rc8_m15_plate PPA-CF 1 67.6 x 18.0 x 11.0 Pins at 49.60
rc8_m15_z42 PPA-CF 1 41.0 x 29.6 x 6.0 drum wrap idler output m1.5 z42 sector
rc8_m15_z24 PPA-CF 1 38.9 x 38.9 x 6.0 reel cup ring gear m1.5 z24
rc8_m2_plate PPA-CF 1 90.1 x 18.0 x 11.0 Pins at 72.10
rc8_m2_z50 PPA-CF 1 61.0 x 39.3 x 6.0 drum REEL output m2 z50 sector
rc8_m2_z22 PPA-CF 1 48.0 x 47.7 x 6.0 reel capstan gear m2 z22
rc8_m25_plate PPA-CF 1 100.6 x 18.0 x 11.0 Pins at 82.60
rc8_m25_big PPA-CF 1 61.5 x 49.0 x 6.0 launch big gear m2.5 z40 sector: teeth 21..26 (tall, relieved)
rc8_m25_drive PPA-CF 1 42.5 x 44.0 x 6.0 launch drive gear m2.5 z26 sector
rc9_hub_lever PPA-CF 1 79.0 x 26.0 x 15.5 drum.coupler_hub slice (shear pin hole) with a lever, string hole at r 60
rc9_outer_lever PPA-CF 1 86.0 x 40.0 x 9.5 drum.coupler slice (shear pin hole) with a lever, string hole at r 60
rc9_shear_pin PLA 3 12.1 x 5.0 x 4.1 drum.shear_pin as built (d4 PLA), printed LYING on its 0.4 mm flat as drum prints it (D44): the shear plane runs across the layers
rc10_magnet_cap PLA 1 20.0 x 20.0 x 6.5 drum.magnet_holder as built: AS5600 magnet cap on the d12 spigot end

# 3. Parts list

## Printed parts

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| drum_bracket_b | drum | PLA | 1 | 25.6 | Drum bracket B: axle, guide rod, wrap idler journal; foot screwed to the reel head spine |
| drum_dog_post | drum | PLA | 1 | 7.1 | Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged) |
| drum_drum_a | drum | PLA | 1 | 103.2 | Drum A: N20 gear 120T m0.8, groove 1 barrel (follower), hex spigot |
| drum_drum_b | drum | PLA | 1 | 82.4 | Drum B: brake cam (spool clutch lever), Gray code bands A B C, wrap segment gear + locking arc |
| drum_follower | drum | PLA | 1 | 6.2 | Follower slider: pin in groove 1, slides along the guide rod; the link takes its motion to the bridge |
| drum_fork | drum | PLA | 1 | 19.9 | Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), lug for the bell crank pin, contact tab |
| drum_index_block | drum | PLA | 1 | 33.2 | SW_INDEX block: column on the motor plate front face (2 x M3 x 10 into the frame inserts), M3 nut contact |
| drum_index_lever | drum | PLA | 1 | 0.9 | SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder) |
| drum_link_eye_1 | drum | PLA | 1 | 3.0 | Link eye pressed on the rod end, M3 screw through the follower lug |
| drum_link_eye_2 | drum | PLA | 1 | 3.0 | Link eye pressed on the rod end, M3 screw through the bell crank input arm |
| drum_magnet_cap | drum | PLA | 1 | 1.9 | AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| drum_n20_cradle | drum | PLA | 1 | 12.1 | N20 cradle on the reel head front plate: pivot + swing screw (hand fallback), SW_N20 contact |
| drum_n20_strap | drum | PLA | 1 | 2.7 | End plate closing the N20 cradle, 2 x M3 x 8 |
| drum_shear_pin | drum | PLA | 6 | 0.2 | Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares |
| drum_sw_lever_0 | drum | PLA | 1 | 0.9 | Code switch lever A: flexure, round nub, M3 x 6 contact screw |
| drum_sw_lever_1 | drum | PLA | 1 | 0.9 | Code switch lever B: flexure, round nub, M3 x 6 contact screw |
| drum_sw_lever_2 | drum | PLA | 1 | 0.9 | Code switch lever C: flexure, round nub, M3 x 6 contact screw |
| drum_switch_block | drum | PLA | 1 | 18.3 | Code switch block on top of the drum: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C) |
| frame_cable_anchor_1 | frame | PLA | 1 | 4.0 | cable-tie anchor (strain relief), 1 x M3 x 12 into the deck |
| frame_cable_anchor_2 | frame | PLA | 1 | 4.0 | cable-tie anchor (strain relief), 1 x M3 x 12 into the deck |
| frame_camera_mount | frame | PLA | 1 | 166.5 | Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck |
| frame_can_stay | frame | PLA | 1 | 28.4 | can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter |
| frame_cheek_r | frame | PLA | 1 | 205.2 | motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems |
| frame_deck_1 | frame | PLA | 1 | 233.6 | deck: M3 inserts for every frame foot; 10 x M3 rover holes (it is its own drill template) |
| frame_deck_2 | frame | PLA | 1 | 304.3 | deck: M3 inserts for every frame foot; 12 x M3 rover holes (it is its own drill template) |
| frame_deck_3 | frame | PLA | 1 | 231.4 | deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template) |
| frame_deck_4 | frame | PLA | 1 | 301.6 | deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template) |
| frame_ebay | frame | PLA | 1 | 175.3 | electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors |
| frame_ebay_esp_clamp | frame | PLA | 1 | 1.6 | ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts |
| frame_ebay_lid | frame | PLA | 1 | 71.4 | bay lid: grille over the BTS7960, 4 x M3 into the corner bosses |
| frame_guard_front | frame | PLA | 1 | 56.8 | front gear guard between the drum gears and the bay, M3 x 10 into the deck |
| frame_guard_hood | frame | PLA | 1 | 14.1 | gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate |
| frame_motor_leg | frame | PLA | 1 | 117.3 | motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate |
| frame_rc832_holder | frame | PLA | 1 | 22.8 | RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10 |
| launch_aim_cap | launch | PLA | 1 | 5.2 | Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles |
| launch_aim_crankpin_clip | launch | PLA | 1 | 0.1 | Printed C clip on the crank pin, inboard of the conrod |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| launch_aim_drag_pad | launch | PLA | 1 | 0.4 | Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket |
| launch_aim_drag_shim | launch | PLA | 1 | 0.1 | Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %) |
| launch_aim_housing | launch | PLA | 1 | 185.4 | Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts |
| launch_aim_idler_clip | launch | PLA | 1 | 0.1 | Printed C clip on the arm pin, holds the idler |
| launch_aim_lugpin_clip | launch | PLA | 1 | 0.1 | Printed C clip on the lug pin, outboard of the conrod |
| launch_carriage | launch | PLA | 1 | 59.6 | Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet |
| launch_carriage_yoke | launch | PLA | 1 | 23.5 | Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62) |
| launch_drive_washer | launch | PLA | 1 | 0.2 | Spacer washer: drive gear hub to the left cheek |
| launch_gear_clip | launch | PLA | 1 | 0.3 | C clip on the big gear pin groove |
| launch_gear_washer | launch | PLA | 1 | 0.5 | Spacer washer: big gear hub to the left cheek |
| launch_guard_l_a | launch | PLA | 1 | 65.0 | Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward |
| launch_guard_l_b1 | launch | PLA | 1 | 4.5 | Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward |
| launch_guard_l_b2 | launch | PLA | 1 | 17.9 | Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward |
| launch_guard_l_c | launch | PLA | 1 | 4.8 | Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward |
| launch_guard_r_a | launch | PLA | 1 | 65.0 | Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward |
| launch_guard_r_b1 | launch | PLA | 1 | 4.5 | Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward |
| launch_guard_r_b2 | launch | PLA | 1 | 21.8 | Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward |
| launch_guard_r_c | launch | PLA | 1 | 4.8 | Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward |
| launch_muzzle_roller | launch | PLA | 1 | 0.5 | Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| launch_muzzle_washer_l | launch | PLA | 1 | 0.0 | Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot |
| launch_muzzle_washer_r | launch | PLA | 1 | 0.0 | Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot |
| launch_rail_front | launch | PLA | 1 | 137.7 | Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie |
| launch_rail_mid | launch | PLA | 1 | 178.7 | Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots |
| launch_rail_rear | launch | PLA | 1 | 115.9 | Rail rear piece: shoe track, walls, flange at the rear end for the reel head (D49) |
| launch_rail_splice_l | launch | PLA | 1 | 0.3 | Top splice strap over the front rail joint (4 x M3 x 10 self tap) |
| launch_rail_splice_r | launch | PLA | 1 | 0.3 | Top splice strap over the front rail joint (4 x M3 x 10 self tap) |
| launch_seat_clip | launch | PLA | 1 | 0.1 | Head cap on the plunger pin (rests on the yoke block) |
| launch_stop_frame_l | launch | PLA | 1 | 32.7 | Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs |
| launch_stop_frame_r | launch | PLA | 1 | 32.7 | Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs |
| launch_stop_spring_l | launch | PLA | 1 | 81.6 | Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall |
| launch_stop_spring_r | launch | PLA | 1 | 81.6 | Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall |
| launch_sw_home | launch | PLA | 1 | 0.8 | SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %) |
| launch_sw_seated | launch | PLA | 1 | 0.8 | SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %) |
| reel_bridge | reel | PLA | 1 | 118.5 | Rail bridge: plate inboard of the main-axis stack with countershafts C and T, the c1' stub, pillars to the u-shaft bushings, cad-drum's shift rail; box arm bolted to rail cheek A. Print on its inboard face |
| reel_cup | reel | PLA | 1 | 12.7 | Wrap cup (Harry v7, D49 axial eyelet at r 26): the line runs straight through the eyelet. Print open end down |
| reel_cup_clip | reel | PLA | 1 | 0.2 | Harry's v7 C-clip on the cup shaft neck in front of the cup |
| reel_head_front | reel | PLA | 1 | 108.4 | Head front plate: bolts the reel head to the rail rear flange, u-shaft T rear bushing, link window. Print flat |
| reel_head_mid | reel | PLA | 1 | 19.3 | Head mid plate: spool shaft inboard bushing. Print flat |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| reel_head_p1 | reel | PLA | 1 | 55.0 | Head plate P1: capstan tube, u-shaft C and G_b bushings, clutch lever post, spool shaft leg. Print flat |
| reel_head_p1b | reel | PLA | 1 | 10.7 | Head plate P1b: capstan tube rear bushing, SW_ARM. Print flat |
| reel_head_p2 | reel | PLA | 1 | 94.1 | Head plate P2: cup shaft and wrap idler rear journals. Print flat |
| reel_head_spine | reel | PLA | 1 | 104.2 | Head spine: inboard wall from the front plate to P2; the drum brackets and plates screw to it. Print flat |
| reel_shroud | reel | PLA | 1 | 7.6 | Spool shroud: keeps loose loops on the spool, open at the top front where the line leaves |
| reel_spool | reel | PLA | 1 | 30.8 | Spool d40 core, d62 flanges, 20 wide: 20 m of 1 mm Dyneema, turns free on two 608s. Print flange down |
| reel_sw_base | reel | PLA | 1 | 7.0 | SW_ARM contact base on P2's front face |
| reel_sw_lever | reel | PLA | 1 | 1.3 | SW_ARM lever: rides the cup gear's raised track (contact closed), drops into its home notch (open) |
| sensors_as5600_carrier | sensors | PLA | 1 | 12.7 | AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine |
| sensors_as5600_strip_bot | sensors | PLA | 1 | 1.0 | AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back |
| sensors_as5600_strip_top | sensors | PLA | 1 | 1.0 | AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back |
| sensors_sw_gap | sensors | PLA | 1 | 9.9 | SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap) |
| sensors_sw_gap_lever | sensors | PLA | 1 | 1.4 | SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam |
| sensors_sw_rail_low | sensors | PLA | 1 | 7.3 | SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point |
| sensors_sw_rail_low_bracket | sensors | PLA | 1 | 17.8 | SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert |
| sensors_sw_rail_low_lever | sensors | PLA | 1 | 1.3 | SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam |
| drum_axle | drum | PPA-CF | 1 | 8.9 | Fixed drum axle d10, pressed into the reel head front plate and bracket B |
| drum_collar | drum | PPA-CF | 1 | 44.0 | Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs) |
| drum_coupler | drum | PPA-CF | 1 | 89.3 | Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring |
| drum_coupler_hub | drum | PPA-CF | 1 | 14.6 | Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole |
| drum_crank | drum | PPA-CF | 1 | 6.9 | Bell crank 1:1 on the bridge shelf: turns the link's motion along the rail into the fork's shift |
| drum_dogring_launch | drum | PPA-CF | 1 | 19.0 | LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side |
| drum_gear_reel | drum | PPA-CF | 1 | 95.9 | REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar; meshes C1 on the rail bridge (D49) |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| drum_idler | drum | PPA-CF | 1 | 8.7 | Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, journals in bracket B and P2 |
| drum_idler_gear | drum | PPA-CF | 1 | 35.8 | Wrap idler output gear 49T m1.5 (D bore) to the cad-reel cup gear 28T: 7 cup turns |
| drum_n20_pinion | drum | PPA-CF | 1 | 0.5 | N20 pinion 10T m0.8, D bore pressed on the N20 shaft |
| drum_reel_washer | drum | PPA-CF | 1 | 0.9 | REEL gear retaining washer on the coupler journal end, 2 x M3 x 6 |
| drum_rod | drum | PPA-CF | 1 | 8.9 | Follower guide rod d10, pressed into the reel head front plate and bracket B |
| drum_shaft | drum | PPA-CF | 1 | 45.2 | Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot) |
| drum_shift_rail | drum | PPA-CF | 1 | 8.2 | Shift rail d10 on the rail bridge (pressed into the bridge plate boss) |
| frame_motor_adapter | frame | PPA-CF | 1 | 302.8 | puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws, 2 inserts for cad-drum's SW_INDEX block; reprint when the motor is measured |
| launch_aim_arm | launch | PPA-CF | 1 | 6.6 | Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove |
| launch_aim_bevel30 | launch | PPA-CF | 1 | 8.8 | Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1 |
| launch_aim_conrod | launch | PPA-CF | 1 | 14.2 | Conrod 156 between pin centres (PPA-CF), printed C clips on both pins |
| launch_aim_crank | launch | PPA-CF | 1 | 35.3 | Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside |
| launch_aim_idler | launch | PPA-CF | 1 | 7.5 | Splitter idler m1.5 z18 face 12 on the arm pin |
| launch_aim_lug | launch | PPA-CF | 1 | 16.9 | Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0) |
| launch_aim_output | launch | PPA-CF | 1 | 26.2 | Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat) |
| launch_aim_sun | launch | PPA-CF | 1 | 18.3 | Splitter sun m1.5 z30 face 12, keyed on the launch sleeve |
| launch_aim_worm | launch | PPA-CF | 1 | 21.3 | Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15 |
| launch_big_gear | launch | PPA-CF | 1 | 177.6 | Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin |
| launch_drive_gear | launch | PPA-CF | 1 | 29.9 | Drive gear m2.5 26T, sprag #1 inside on an integral key rib |
| launch_gear_pin | launch | PPA-CF | 1 | 5.4 | Snap pin d10 for the big gear, C clip; prints lying on its flat |
| launch_seat_plunger | launch | PPA-CF | 1 | 1.0 | SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top |
| launch_seat_pushrod | launch | PPA-CF | 1 | 0.2 | SW_SEATED pushrod d3.2 with the stem slot |
| launch_sleeve | launch | PPA-CF | 1 | 49.3 | Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals |
| launch_yoke_peg_l | launch | PPA-CF | 1 | 1.2 | Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| launch_yoke_peg_r | launch | PPA-CF | 1 | 1.2 | Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred |
| reel_b1g | reel | PPA-CF | 1 | 2.6 | Bevel B1 gear m1.5 24T (2:1 up to u-shaft C) |
| reel_b1p | reel | PPA-CF | 1 | 1.2 | Bevel B1 pinion m1.5 12T on the front end of u-shaft C |
| reel_b2g | reel | PPA-CF | 1 | 0.9 | Bevel B2 m1 20T on countershaft T |
| reel_b2p | reel | PPA-CF | 1 | 1.3 | Bevel B2 m1 20T on the front end of u-shaft T |
| reel_b3a | reel | PPA-CF | 1 | 1.3 | Bevel B3 m1 20T on the rear end of u-shaft T |
| reel_b3b | reel | PPA-CF | 1 | 1.4 | Bevel B3 m1 20T on the spool shaft |
| reel_c | reel | PPA-CF | 1 | 17.4 | Capstan tube C (PPA-CF d20): key rib in the drum, G_c on its flat, journals in P1 and P1b. Print standing on its rear end |
| reel_c1 | reel | PPA-CF | 1 | 93.7 | C1 m2 49T: meshes the REEL output gear 50T across the pivot |
| reel_c1p | reel | PPA-CF | 1 | 27.8 | c1' m1.5 46T, free on its stub: driven by the bus gear 50T, drives the swing idler |
| reel_cs_c | reel | PPA-CF | 1 | 11.4 | Countershaft C (PPA-CF): sprag #2, C1 and the B1 bevel. Print standing |
| reel_cs_t | reel | PPA-CF | 1 | 2.8 | Countershaft T (PPA-CF): T1 and the B2 bevel |
| reel_d | reel | PPA-CF | 1 | 7.6 | Clutch disk D on the spool shaft flat: its friction ring presses the spool's outboard face. Print ring up |
| reel_drum | reel | PPA-CF | 1 | 45.0 | Capstan drum dia 40 x 40 (Harry v7), keyed on the tube. Print flange down |
| reel_ga | reel | PPA-CF | 1 | 9.6 | G_a m1.5 26T on the rear end of u-shaft C |
| reel_gb | reel | PPA-CF | 1 | 15.1 | G_b m1.5 32T idler on its stub in P1 |
| reel_gc | reel | PPA-CF | 1 | 5.3 | Capstan gear G_c m1.5 23T on the tube flat |
| reel_gi | reel | PPA-CF | 1 | 11.4 | Swing idler m1.5 30T |
| reel_gk | reel | PPA-CF | 1 | 15.0 | Cup gear G_k m1.5 28T: driven by the drum's wrap idler 49T (7 cup turns per wrap); the raised track on its rear face holds SW_ARM closed, the notch at home lets it open. Print track face down |
| reel_k | reel | PPA-CF | 1 | 10.2 | Cup shaft K (PPA-CF d10): runs inside the tube, cup in front, G_k behind P1b, journal in P2. Print standing on its rear end |
| reel_lever | reel | PPA-CF | 1 | 6.9 | Clutch lever (PPA-CF): the drum's brake cam presses its upper arm; the lower arm is the clutch spring leaf, its FIRM screw squeezes the spool. Print flat |
| reel_s | reel | PPA-CF | 1 | 5.7 | Spool shaft S (PPA-CF d8): B3 bevel and the clutch disk on its flats, journals in the mid plate and the P1 leg. Print lying on its 1 mm flat |
| reel_sprag_hub | reel | PPA-CF | 1 | 12.2 | Sprag #2 hub on the bridge: holds the CSK20PP capstan backstop |
| reel_stub | reel | PPA-CF | 1 | 1.1 | c1' stub axle (PPA-CF d8, pressed into the bridge) |
| reel_swing_arm | reel | PPA-CF | 1 | 3.2 | Swing arm on c1's hub: carries the idler into mesh with T1 on forward wiper turns |
| reel_t | reel | PPA-CF | 1 | 1.1 | Thrust ring: the lever's FIRM screw pushes it onto the inboard 608 inner race (clutch squeeze) |
| reel_t1 | reel | PPA-CF | 1 | 4.6 | T1 m1.5 20T: take-up at 2.5 x wiper |

| part | subsystem | material | qty | mass g | notes |
|---|---|---|---|---|---|
| reel_uc | reel | PPA-CF | 1 | 23.7 | u-shaft C (PPA-CF d10, 244 long): along the rail from the B1 bevel to G_a. Print standing (244 of the 260 mm height); an 8 mm steel rod with printed sleeves is the stiff fallback |
| reel_ut | reel | PPA-CF | 1 | 6.5 | u-shaft T (PPA-CF d8): along the rail from the B2 bevel to the B3 bevel. Print standing |

Printed mass: PLA 4011 g (1344.9 m of filament); PPA-CF 1456 g (484.4 m of filament).

## Bought parts

| item | type | count | used by |
|---|---|---|---|
| 608ZZ 8x22x7 | bearings | 4 | launch_aim_brg_1, launch_aim_brg_2, reel_brg_1, reel_brg_2 |
| 693ZZ 3x8x4 | bearings | 1 | launch_muzzle_roller_brg |
| Adafruit AS5600 STEMMA QT PID 6357 | bearings | 1 | sensors_as5600 |
| CSK20PP 20x47x14 sprag #1 | bearings | 1 | launch_sprag |
| CSK20PP one-way bearing 20x47x14 | bearings | 1 | reel_sprag |
| M3 nut | fasteners | 6 | launch_sw_seated_nut, launch_sw_home_nut, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_sw_rail_low_nut, sensors_sw_gap_nut |
| M3 nut (jam) | fasteners | 2 | sensors_sw_rail_low_screw, sensors_sw_gap_screw |
| M3 nyloc nut | fasteners | 1 | launch_muzzle_roller_axle |
| M3x10 self tap | fasteners | 2 | launch_guard_l_b2, launch_guard_r_b2 |
| M3x10 self tap (lever clamp) | fasteners | 6 | reel_sw_lever, sensors_sw_rail_low_lever, sensors_sw_gap_lever |
| M3x10 SHCS | fasteners | 4 | drum_index_block, sensors_sw_rail_low_screw, sensors_sw_gap_screw |
| M3x10 SHCS (base to P2) | fasteners | 4 | reel_sw_base |
| M3x10 SHCS (foot into the housing) | fasteners | 1 | sensors_sw_rail_low_bracket |
| M3x10 SHCS + 2 M3 nuts | fasteners | 1 | reel_sw_hw |
| M3x10 SHCS + M3 nut | fasteners | 1 | reel_firm_screw |
| M3x12 self tap | fasteners | 17 | launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c |
| M3x12 SHCS | fasteners | 4 | reel_head_mid, sensors_as5600_strip_top, sensors_as5600_strip_bot |
| M3x12 SHCS (arm into the right cheek) | fasteners | 2 | sensors_sw_gap |
| M3x12 SHCS (posts) | fasteners | 2 | reel_head_p1b |
| M3x12 SHCS (spine) | fasteners | 7 | reel_head_front, reel_head_p1, reel_head_p2 |
| M3x12 SHCS (through the spine) | fasteners | 4 | drum_bracket_b, drum_switch_block |
| M3x16 SHCS | fasteners | 5 | launch_muzzle_roller_axle, reel_sprag_hub |
| M3x16 SHCS (arm to cheek A) | fasteners | 2 | reel_bridge |
| M3x16 SHCS (deck screw) | fasteners | 1 | sensors_sw_rail_low_bracket |
| M3x16 SHCS (into the housing spine) | fasteners | 2 | sensors_as5600_carrier |
| M3x16 SHCS (into the rail flange inserts) | fasteners | 4 | reel_head_front |
| M3x16 SHCS (pivot, tapped into the shelf) | fasteners | 1 | drum_crank |
| M3x16 SHCS + M3 nut | fasteners | 1 | reel_light_screw |
| M3x16 SHCS + nyloc nut | fasteners | 2 | drum_link_eye_1, drum_link_eye_2 |

| item | type | count | used by |
|---|---|---|---|
| M3x16 SHCS + washer | fasteners | 1 | reel_lever_pin |
| M3x6 SHCS | fasteners | 2 | launch_sw_seated_screw, launch_sw_home_screw |
| M3x8 self tap (base to bracket) | fasteners | 4 | sensors_sw_rail_low |
| M3x8 SHCS (tab to the P1 leg) | fasteners | 1 | reel_shroud |
| M8 hex nut ISO 4032 | fasteners | 1 | launch_aim_nut |
| M8x25 hex bolt ISO 4017 | fasteners | 1 | launch_aim_bolt |
| STEMMA QT JST SH 4 pin cable | fasteners | 1 | sensors_as5600 |
| (part of the wiper motor) | other | 1 | frame_wiper_shaft |
| 12 V car wiper motor (self-locking worm), 2-wire rebuild | other | 1 | frame_wiper_motor |
| 8 mm steel rod, 26 mm long | other | 1 | reel_gb_axle |
| 8 mm steel rod, cut to length (see note), ends deburred | other | 2 | launch_anchor_pin_l, launch_anchor_pin_r |
| 8 mm steel rod, cut to the length on the print sheet | other | 1 | drum_link |
| BTS7960 43 A module (IBT-2) | other | 1 | frame_pcb_bts7960 |
| Century C-243 (5/8 x 6-1/2 in x 0.054 in wire) extension spring | other | 2 | launch_spring_l_hook_a, launch_spring_r_hook_a |
| Dyneema braid 1 mm (LINE) | other | 21 | launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8, launch_line_a9, launch_line_a10, launch_line_a11, launch_line_a12, launch_line_a13, launch_line_a14, launch_line_a15, launch_line_a16, launch_line_a17, launch_line_a18, launch_line_a19, launch_line_a20 |
| Espressif ESP32-S3-DevKitC-1-N8 | other | 1 | frame_pcb_esp32 |
| HX711 load cell amplifier (optional) | other | 1 | frame_pcb_hx711 |
| inline ATO fuse holder + 20 A fuse | other | 1 | frame_fuse_20a |
| KEDGE v30 dart (reference, built separately) | other | 1 | launch_dart |
| launch_line_p (bought, no bought_ref) | other | 1 | launch_line_p |
| launch_spring_l_0 (bought, no bought_ref) | other | 1 | launch_spring_l_0 |
| launch_spring_l_1 (bought, no bought_ref) | other | 1 | launch_spring_l_1 |
| launch_spring_l_2 (bought, no bought_ref) | other | 1 | launch_spring_l_2 |
| launch_spring_l_3 (bought, no bought_ref) | other | 1 | launch_spring_l_3 |
| launch_spring_l_4 (bought, no bought_ref) | other | 1 | launch_spring_l_4 |
| launch_spring_l_hook_c (bought, no bought_ref) | other | 1 | launch_spring_l_hook_c |
| launch_spring_r_0 (bought, no bought_ref) | other | 1 | launch_spring_r_0 |
| launch_spring_r_1 (bought, no bought_ref) | other | 1 | launch_spring_r_1 |
| launch_spring_r_2 (bought, no bought_ref) | other | 1 | launch_spring_r_2 |

| item | type | count | used by |
|---|---|---|---|
| launch_spring_r_3 (bought, no bought_ref) | other | 1 | launch_spring_r_3 |
| launch_spring_r_4 (bought, no bought_ref) | other | 1 | launch_spring_r_4 |
| launch_spring_r_hook_c (bought, no bought_ref) | other | 1 | launch_spring_r_hook_c |
| MP1584EN buck module, set to 9.0 V | other | 1 | frame_pcb_mp1584 |
| N20 right-angle self-locking worm gearmotor 12 V 27 rpm (have) | other | 1 | drum_n20 |
| perfboard 55 x 31 (dividers, pull-downs, 3V3 and GND buses), 2 x M3 x 10 on wall bosses | other | 1 | frame_perfboard |
| Pololu DRV8833 2130 (headers off) | other | 1 | frame_pcb_drv8833 |
| Razer Kiyo X webcam | other | 1 | frame_kiyo_x |
| RC832 5.8 GHz receiver (80 x 65 x 15) | other | 1 | frame_rc832 |
| sensors_as5600_col_screws (bought, no bought_ref) | other | 1 | sensors_as5600_col_screws |
| sensors_as5600_nut_bot (bought, no bought_ref) | other | 1 | sensors_as5600_nut_bot |
| sensors_as5600_nut_top (bought, no bought_ref) | other | 1 | sensors_as5600_nut_top |
| sensors_as5600_screw_bot (bought, no bought_ref) | other | 1 | sensors_as5600_screw_bot |
| sensors_as5600_screw_top (bought, no bought_ref) | other | 1 | sensors_as5600_screw_top |
| sensors_sw_gap_jam (bought, no bought_ref) | other | 1 | sensors_sw_gap_jam |
| sensors_sw_rail_low_jam (bought, no bought_ref) | other | 1 | sensors_sw_rail_low_jam |

## 4. Print list

150 printed parts on 19 plates. The 3MF plates and STL files are in the print pack.

| part | material | qty | print size mm | supports |
|---|---|---|---|---|
| frame_deck_2 | PLA | 1 | 222 x 212 x 8 | no |
| frame_deck_4 | PLA | 1 | 214 x 212 x 8 | no |
| frame_deck_1 | PLA | 1 | 222 x 169 x 8 | no |
| frame_deck_3 | PLA | 1 | 214 x 169 x 8 | no |
| frame_cheek_r | PLA | 1 | 215 x 220 x 30 | yes, tree supports from the plate (1084 mm2 over 45 deg) |
| frame_camera_mount | PLA | 1 | 58 x 72 x 236 | yes, tree supports from the plate (672 mm2 over 45 deg) |
| launch_aim_housing | PLA | 1 | 176 x 59 x 183 | yes, tree supports from the plate (761 mm2 over 45 deg) |
| launch_rail_mid | PLA | 1 | 92 x 68 x 170 | yes, tree supports from the plate (2356 mm2 over 45 deg) |
| frame_motor_leg | PLA | 1 | 49 x 69 x 187 | yes, tree supports from the plate (222 mm2 over 45 deg) |
| launch_rail_front | PLA | 1 | 132 x 128 x 48 | yes, tree supports from the plate (7890 mm2 over 45 deg) |

| part | material | qty | print size mm | supports |
|---|---|---|---|---|
| frame_ebay | PLA | 1 | 124 x 124 x 58 | yes, tree supports from the plate (879 mm2 over 45 deg) |
| drum_drum_a | PLA | 1 | 97 x 97 x 43 | yes, tree supports from the plate (719 mm2 over 45 deg) |
| launch_rail_rear | PLA | 1 | 151 x 68 x 40 | yes, tree supports from the plate (3676 mm2 over 45 deg) |
| reel_bridge | PLA | 1 | 220 x 192 x 88 | yes, tree supports from the plate (686 mm2 over 45 deg) |
| reel_head_front | PLA | 1 | 188 x 186 x 6 | no |
| reel_head_spine | PLA | 1 | 131 x 188 x 6 | no |
| drum_drum_b | PLA | 1 | 77 x 77 x 41 | yes, tree supports from the plate (3126 mm2 over 45 deg) |
| reel_head_p2 | PLA | 1 | 104 x 182 x 6 | no |
| launch_stop_spring_l | PLA | 1 | 155 x 52 x 14 | no |
| launch_stop_spring_r | PLA | 1 | 155 x 52 x 14 | no |
| frame_ebay_lid | PLA | 1 | 124 x 124 x 7 | no |
| reel_head_p1 | PLA | 1 | 100 x 177 x 37 | no |
| launch_guard_l_a | PLA | 1 | 250 x 49 x 40 | no |
| launch_guard_r_a | PLA | 1 | 250 x 49 x 40 | no |
| launch_carriage | PLA | 1 | 205 x 240 x 43 | yes, tree supports from the plate (6453 mm2 over 45 deg) |
| frame_guard_front | PLA | 1 | 155 x 104 x 16 | no |
| drum_index_block | PLA | 1 | 25 x 68 x 40 | no |
| frame_can_stay | PLA | 1 | 55 x 55 x 71 | yes, tree supports from the plate (906 mm2 over 45 deg) |
| reel_spool | PLA | 1 | 62 x 62 x 22 | yes, tree supports from the plate (1791 mm2 over 45 deg) |
| launch_stop_frame_l | PLA | 1 | 175 x 53 x 55 | yes, tree supports from the plate (541 mm2 over 45 deg) |
| launch_stop_frame_r | PLA | 1 | 175 x 53 x 55 | yes, tree supports from the plate (541 mm2 over 45 deg) |
| drum_bracket_b | PLA | 1 | 76 x 109 x 6 | no |
| drum_switch_block | PLA | 1 | 26 x 66 x 21 | yes, tree supports from the plate (216 mm2 over 45 deg) |
| launch_carriage_yoke | PLA | 1 | 43 x 127 x 28 | yes, tree supports from the plate (1406 mm2 over 45 deg) |
| frame_rc832_holder | PLA | 1 | 97 x 22 x 40 | no |
| reel_head_mid | PLA | 1 | 54 x 58 x 6 | yes, tree supports from the plate (143 mm2 over 45 deg) |
| launch_guard_r_b2 | PLA | 1 | 145 x 32 x 40 | no |
| drum_fork | PLA | 1 | 86 x 68 x 51 | yes, tree supports from the plate (1227 mm2 over 45 deg) |
| sensors_sw_rail_low_bracket | PLA | 1 | 31 x 48 x 74 | no |
| launch_guard_l_b2 | PLA | 1 | 145 x 32 x 40 | no |
| sensors_as5600_carrier | PLA | 1 | 34 x 68 x 11 | no |
| frame_guard_hood | PLA | 1 | 122 x 56 x 14 | no |
| reel_cup | PLA | 1 | 48 x 53 x 18 | yes, tree supports from the plate (1210 mm2 over 45 deg) |

| part | material | qty | print size mm | supports |
|---|---|---|---|---|
| drum_n20_cradle | PLA | 1 | 54 x 32 x 20 | yes, tree supports from the plate (114 mm2 over 45 deg) |
| reel_head_p1b | PLA | 1 | 38 x 44 x 16 | no |
| sensors_sw_gap | PLA | 1 | 63 x 43 x 12 | no |
| reel_shroud | PLA | 1 | 70 x 66 x 30 | no |
| sensors_sw_rail_low | PLA | 1 | 48 x 35 x 12 | no |
| drum_dog_post | PLA | 1 | 22 x 65 x 48 | no |
| reel_sw_base | PLA | 1 | 46 x 35 x 12 | no |
| drum_follower | PLA | 1 | 26 x 39 x 17 | yes, tree supports from the plate (260 mm2 over 45 deg) |
| launch_aim_cap | PLA | 1 | 46 x 14 x 19 | no |
| launch_guard_l_c | PLA | 1 | 32 x 54 x 26 | no |
| launch_guard_r_c | PLA | 1 | 32 x 54 x 26 | no |
| launch_guard_l_b1 | PLA | 1 | 12 x 49 x 40 | no |
| launch_guard_r_b1 | PLA | 1 | 12 x 49 x 40 | no |
| frame_cable_anchor_1 | PLA | 1 | 16 x 24 x 10 | yes, tree supports from the plate (141 mm2 over 45 deg) |
| frame_cable_anchor_2 | PLA | 1 | 24 x 16 x 10 | yes, tree supports from the plate (141 mm2 over 45 deg) |
| drum_link_eye_1 | PLA | 1 | 13 x 13 x 23 | no |
| drum_link_eye_2 | PLA | 1 | 13 x 13 x 23 | no |
| drum_n20_strap | PLA | 1 | 54 x 18 x 3 | no |
| drum_magnet_cap | PLA | 1 | 20 x 20 x 6 | no |
| frame_ebay_esp_clamp | PLA | 1 | 43 x 8 x 4 | no |
| sensors_sw_gap_lever | PLA | 1 | 40 x 9 x 8 | no |
| sensors_sw_rail_low_lever | PLA | 1 | 44 x 5 x 8 | no |
| reel_sw_lever | PLA | 1 | 42 x 5 x 8 | no |
| sensors_as5600_strip_top | PLA | 1 | 25 x 12 x 3 | no |
| sensors_as5600_strip_bot | PLA | 1 | 25 x 12 x 3 | no |
| drum_index_lever | PLA | 1 | 13 x 48 x 6 | no |
| drum_sw_lever_0 | PLA | 1 | 14 x 48 x 6 | no |
| drum_sw_lever_1 | PLA | 1 | 14 x 48 x 6 | no |
| drum_sw_lever_2 | PLA | 1 | 14 x 48 x 6 | no |
| launch_sw_seated | PLA | 1 | 42 x 8 x 6 | no |
| launch_sw_home | PLA | 1 | 42 x 8 x 6 | no |
| launch_muzzle_roller | PLA | 1 | 14 x 14 x 5 | no |
| launch_gear_washer | PLA | 1 | 18 x 18 x 2 | no |
| launch_aim_drag_pad | PLA | 1 | 9 x 24 x 4 | no |
| launch_gear_clip | PLA | 1 | 17 x 18 x 1 | no |
| launch_rail_splice_l | PLA | 1 | 22 x 5 x 3 | no |
| launch_rail_splice_r | PLA | 1 | 22 x 5 x 3 | no |
| launch_drive_washer | PLA | 1 | 32 x 32 x 1 | no |
| drum_shear_pin | PLA | 6 | 12 x 5 x 4 | no |
| reel_cup_clip | PLA | 1 | 10 x 10 x 2 | no |

| part | material | qty | print size mm | supports |
|---|---|---|---|---|
| launch_aim_drag_shim | PLA | 1 | 9 x 7 x 4 | no |
| launch_aim_idler_clip | PLA | 1 | 11 x 10 x 1 | no |
| launch_aim_crankpin_clip | PLA | 1 | 11 x 10 x 1 | no |
| launch_aim_lugpin_clip | PLA | 1 | 11 x 10 x 1 | no |
| launch_seat_clip | PLA | 1 | 6 x 6 x 2 | no |
| launch_muzzle_washer_l | PLA | 1 | 6 x 6 x 1 | no |
| launch_muzzle_washer_r | PLA | 1 | 6 x 6 x 1 | no |
| frame_motor_adapter | PPA-CF | 1 | 194 x 194 x 16 | yes, tree supports from the plate (5484 mm2 over 45 deg) |
| launch_big_gear | PPA-CF | 1 | 105 x 105 x 27 | yes, tree supports from the plate (639 mm2 over 45 deg) |
| drum_gear_reel | PPA-CF | 1 | 104 x 104 x 19 | no |
| reel_c1 | PPA-CF | 1 | 102 x 102 x 12 | no |
| drum_coupler | PPA-CF | 1 | 78 x 78 x 62 | yes, tree supports from the plate (1998 mm2 over 45 deg) |
| drum_shaft | PPA-CF | 1 | 170 x 25 x 22 | yes, tree supports from the plate (922 mm2 over 45 deg) |
| reel_drum | PPA-CF | 1 | 46 x 46 x 43 | yes, tree supports from the plate (251 mm2 over 45 deg) |
| launch_sleeve | PPA-CF | 1 | 64 x 64 x 85 | yes, tree supports from the plate (2907 mm2 over 45 deg) |
| drum_collar | PPA-CF | 1 | 60 x 60 x 26 | yes, tree supports from the plate (2694 mm2 over 45 deg) |
| launch_aim_crank | PPA-CF | 1 | 62 x 60 x 38 | yes, tree supports from the plate (985 mm2 over 45 deg) |
| drum_idler_gear | PPA-CF | 1 | 76 x 76 x 7 | no |
| launch_drive_gear | PPA-CF | 1 | 70 x 70 x 29 | yes, tree supports from the plate (2457 mm2 over 45 deg) |
| reel_c1p | PPA-CF | 1 | 72 x 72 x 10 | no |
| launch_aim_output | PPA-CF | 1 | 48 x 48 x 60 | no |
| reel_uc | PPA-CF | 1 | 10 x 10 x 244 | no |
| launch_aim_worm | PPA-CF | 1 | 28 x 28 x 54 | yes, tree supports from the plate (2779 mm2 over 45 deg) |
| drum_dogring_launch | PPA-CF | 1 | 62 x 62 x 10 | yes, tree supports from the plate (123 mm2 over 45 deg) |
| launch_aim_sun | PPA-CF | 1 | 48 x 48 x 12 | no |
| reel_c | PPA-CF | 1 | 23 x 20 x 58 | no |
| launch_aim_lug | PPA-CF | 1 | 30 x 58 x 28 | yes, tree supports from the plate (995 mm2 over 45 deg) |
| reel_gb | PPA-CF | 1 | 51 x 51 x 7 | no |
| reel_gk | PPA-CF | 1 | 45 x 45 x 12 | yes, tree supports from the plate (1031 mm2 over 45 deg) |
| drum_coupler_hub | PPA-CF | 1 | 26 x 26 x 32 | no |
| launch_aim_conrod | PPA-CF | 1 | 173 x 17 x 6 | no |
| reel_cs_c | PPA-CF | 1 | 23 x 20 x 40 | yes, tree supports from the plate (182 mm2 over 45 deg) |
| reel_sprag_hub | PPA-CF | 1 | 56 x 56 x 14 | no |

| part | material | qty | print size mm | supports |
|---|---|---|---|---|
| reel_gi | PPA-CF | 1 | 48 x 48 x 6 | no |
| reel_k | PPA-CF | 1 | 10 x 10 x 109 | no |
| reel_ga | PPA-CF | 1 | 42 x 42 x 7 | no |
| drum_axle | PPA-CF | 1 | 95 x 10 x 9 | no |
| drum_rod | PPA-CF | 1 | 95 x 10 x 9 | no |
| launch_aim_bevel30 | PPA-CF | 1 | 53 x 52 x 10 | yes, tree supports from the plate (590 mm2 over 45 deg) |
| drum_idler | PPA-CF | 1 | 22 x 19 x 54 | yes, tree supports from the plate (331 mm2 over 45 deg) |
| drum_shift_rail | PPA-CF | 1 | 88 x 10 x 9 | no |
| reel_d | PPA-CF | 1 | 53 x 53 x 3 | yes, tree supports from the plate (1050 mm2 over 45 deg) |
| launch_aim_idler | PPA-CF | 1 | 30 x 30 x 12 | no |
| drum_crank | PPA-CF | 1 | 56 x 58 x 10 | no |
| reel_lever | PPA-CF | 1 | 67 x 66 x 8 | no |
| launch_aim_arm | PPA-CF | 1 | 64 x 41 x 19 | no |
| reel_ut | PPA-CF | 1 | 8 x 8 x 104 | no |
| reel_s | PPA-CF | 1 | 99 x 8 x 7 | no |
| launch_gear_pin | PPA-CF | 1 | 53 x 16 x 12 | no |
| reel_gc | PPA-CF | 1 | 37 x 37 x 7 | no |
| reel_t1 | PPA-CF | 1 | 33 x 33 x 6 | no |
| reel_swing_arm | PPA-CF | 1 | 73 x 19 x 10 | no |
| reel_cs_t | PPA-CF | 1 | 12 x 12 x 24 | no |
| reel_b1g | PPA-CF | 1 | 39 x 39 x 6 | yes, tree supports from the plate (679 mm2 over 45 deg) |
| reel_b3b | PPA-CF | 1 | 22 x 22 x 9 | no |
| reel_b2p | PPA-CF | 1 | 22 x 22 x 8 | yes, tree supports from the plate (149 mm2 over 45 deg) |
| reel_b3a | PPA-CF | 1 | 22 x 22 x 8 | no |
| launch_yoke_peg_l | PPA-CF | 1 | 20 x 8 x 7 | no |
| launch_yoke_peg_r | PPA-CF | 1 | 20 x 8 x 7 | no |
| reel_b1p | PPA-CF | 1 | 21 x 21 x 11 | no |
| reel_stub | PPA-CF | 1 | 20 x 8 x 7 | no |
| reel_t | PPA-CF | 1 | 18 x 18 x 6 | no |
| launch_seat_plunger | PPA-CF | 1 | 30 x 15 x 20 | no |
| drum_reel_washer | PPA-CF | 1 | 35 x 35 x 1 | no |
| reel_b2g | PPA-CF | 1 | 22 x 22 x 7 | yes, tree supports from the plate (149 mm2 over 45 deg) |
| drum_n20_pinion | PPA-CF | 1 | 10 x 9 x 8 | no |
| launch_seat_pushrod | PPA-CF | 1 | 16 x 5 x 4 | no |

plate_1_PLA.3mf: drum_link_eye_1, drum_link_eye_2, drum_n20_cradle, launch_carriage, launch_guard_l_c, launch_guard_r_c, sensors_sw_rail_low.

plate_2_PLA.3mf: drum_index_block, drum_magnet_cap, drum_switch_block, frame_cheek_r, frame_rc832_holder.

plate_3_PLA.3mf: drum_index_lever, drum_n20_strap, drum_sw_lever_2, frame_cable_anchor_1, frame_deck_2, launch_aim_cap.

plate_4_PLA.3mf: drum_follower, frame_cable_anchor_2, frame_deck_4, launch_aim_drag_pad, sensors_as5600_strip_bot, sensors_as5600_strip_top.

plate_5_PLA.3mf: drum_dog_post, reel_bridge, reel_cup, reel_head_p1b, reel_sw_base, sensors_sw_rail_low_bracket.

plate_6_PLA.3mf: drum_shear_pin__c1, drum_shear_pin__c2, drum_shear_pin__c3, drum_shear_pin__c4, drum_shear_pin__c5, frame_deck_1, launch_aim_crankpin_clip, launch_aim_lugpin_clip, launch_gear_clip, launch_gear_washer, launch_guard_r_b1, launch_rail_rear, launch_sw_home, launch_sw_seated, reel_cup_clip, reel_shroud.

plate_7_PLA.3mf: frame_deck_3, launch_guard_l_b2, launch_guard_r_b2, launch_rail_mid.

plate_8_PLA.3mf: drum_shear_pin__c6, frame_motor_leg, launch_aim_idler_clip, launch_guard_l_b1, launch_muzzle_washer_l, launch_muzzle_washer_r, launch_stop_frame_l, launch_stop_spring_l, reel_head_front.

plate_9_PLA.3mf: launch_stop_frame_r, reel_head_mid, reel_head_p2, reel_head_spine.

plate_10_PLA.3mf: launch_aim_housing, launch_carriage_yoke, launch_rail_front, reel_head_p1, reel_spool, sensors_sw_gap_lever.

plate_11_PLA.3mf: drum_drum_a, drum_fork, drum_sw_lever_0, drum_sw_lever_1, frame_ebay, frame_ebay_esp_clamp, frame_guard_front, launch_aim_drag_shim, launch_drive_washer, launch_muzzle_roller, launch_seat_clip, sensors_as5600_carrier.

plate_12_PLA.3mf: drum_bracket_b, frame_camera_mount, frame_can_stay, frame_ebay_lid, frame_guard_hood, launch_rail_splice_l, launch_rail_splice_r, launch_stop_spring_r, reel_sw_lever, sensors_sw_gap, sensors_sw_rail_low_lever.

plate_13_PLA.3mf: drum_drum_b.

plate_14_PLA.3mf: launch_guard_l_a.

plate_15_PLA.3mf: launch_guard_r_a.

plate_16_PPA-CF.3mf: drum_n20_pinion, drum_reel_washer, drum_rod, frame_motor_adapter, launch_aim_arm, reel_b1g, reel_b1p, reel_b2g, reel_b3a, reel_drum, reel_ga, reel_gc, reel_gk.

plate_17_PPA-CF.3mf: drum_coupler, drum_coupler_hub, drum_gear_reel, drum_idler, drum_shift_rail, launch_aim_bevel30, launch_big_gear, launch_gear_pin, launch_seat_pushrod, reel_b2p, reel_b3b, reel_c, reel_c1, reel_cs_c, reel_d, reel_gb, reel_s, reel_ut.

plate_18_PPA-CF.3mf: drum_axle, drum_crank, drum_dogring_launch, drum_idler_gear, drum_shaft, launch_aim_crank, launch_aim_idler, launch_aim_lug, launch_aim_worm, launch_drive_gear, launch_seat_plunger, launch_sleeve, launch_yoke_peg_l, launch_yoke_peg_r, reel_c1p, reel_cs_t, reel_k, reel_lever, reel_sprag_hub, reel_stub, reel_t1, reel_uc.

plate_19_PPA-CF.3mf: drum_collar, launch_aim_conrod, launch_aim_output, launch_aim_sun, reel_gi, reel_swing_arm, reel_t.

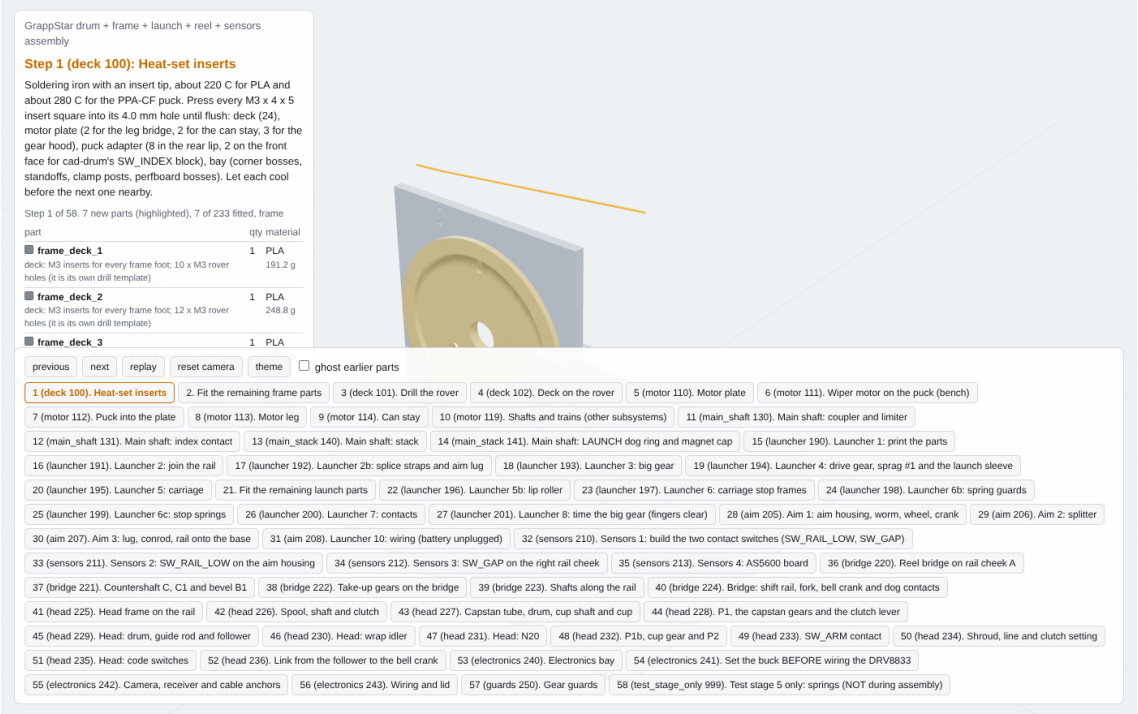
Print settings are in PRINT_SHEET.pdf. Use a hardened nozzle for PPA-CF and dry it first.

# 5. Assembly steps

## Step 1 (deck 100). Heat-set inserts

Battery unplugged. Soldering iron with an insert tip, about 220 C for PLA and about 280 C for the PPA-CF puck. Press every M3 x 4 x 5 insert square into its 4.0 mm hole until flush: deck (24), motor plate (2 for the leg bridge, 2 for the can stay, 3 for the gear hood), puck adapter (8 in the rear lip, 2 on the front face for cad-drum's SW_INDEX block), bay (corner bosses, standoffs, clamp posts, perfboard bosses). Let each cool before the next one nearby.

Parts: frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4, frame_cheek_r, frame_motor_adapter, frame_ebay.




## Step 2. Fit the remaining frame parts

Parts without an explicit assembly step: frame_perfboard.

Parts: frame_perfboard.

## Step 3 (deck 101). Drill the rover

Lay the deck on the rover top plate where it will live (launch direction forward). Mark through the 45 perimeter holes with a 3 mm drill or a scriber (or tape down the 1:1 PDF template from cad/parts/_frame_tests/drill_template.py), take the deck off, drill 3.4 mm. Deburr.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4.

## Step 4 (deck 102). Deck on the rover

Slide the tiles together on the dovetails (from above). M3 x 25 SHCS down through every perimeter hole, washer and nyloc under the rover plate. Snug, not crushed. Do this before the other subsystems put their feet down: some rover bolts sit under their feet.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4.

## Step 5 (motor 110). Motor plate

Hold the motor plate upright (front face, the side without the bridge, towards +y) with its bridge against the top of the leg. 2 x M3 x 12 along y through the leg's long counterbores into the bridge inserts.

Also handle (fitted earlier): frame_cheek_r.

### Step 6 (motor 111). Wiper motor on the puck (bench)

Battery unplugged. Put the M6 bolts through the puck slots from the front (heads sit in the slot counterbores), into the motor bosses (threaded bosses, or a nut behind each boss ear). Slide the motor so its shaft is centred in the bore (a 10 mm drill shank in the coupler bore helps), then tighten.

Parts: frame_wiper_motor, frame_wiper_shaft.
Also handle (fitted earlier): frame_motor_adapter.

### Step 7 (motor 112). Puck into the plate

Feed the puck into the plate window from behind (the motor side) until its lip sits on the plate's rear face. From the front, M3 x 12 through the plate into the lip inserts (8 of the 8 rim holes have both plate and lip; use those).

Also handle (fitted earlier): frame_motor_adapter.

### Step 8 (motor 113). Motor leg

Stand the leg on the deck in the corner between the reel and the bay, foot along y. 4 x M3 x 10 down through the foot into the deck inserts.

Parts: frame_motor_leg.

### Step 9 (motor 114). Can stay

Post against the plate's rear face, 2 x M3 x 12 into the plate inserts. Cable tie or hose clamp round the motor can through the two cradle slots, firm but not crushing.

Parts: frame_can_stay.

### Step 10 (motor 119). Shafts and trains (other subsystems)

Battery unplugged. Now cad-drum (coupler on the wiper shaft, SW_INDEX block on this plate's front face inserts: drum step "Main shaft: index contact"), cad-launch (rail and aim housing) and then the rail bridge and reel head (cad-reel, cad-drum: D49, they ride the rail) go in. Follow their steps, then come back. Leave a flex loop for the head's wires across the pivot (about 10 conductors, 41 deg of pitch).

### Step 11 (main_shaft 130). Main shaft: coupler and limiter

Push the coupler hub onto the wiper output shaft (flats in line) until it stops on the end of the flats; its lower end goes into the puck bore, just clear of the motor's output boss. Drop the M8 nut into the hub pocket onto the thread (it sits about 8 mm down the pocket) and tighten it with a 13 mm socket (1/4 inch drive). Slide the outer coupler over the hub, line up the holes, push a shear pin in (head outside, its flat facing the outboard end) and put a piece of tape round the coupler over the head. Print the pins lying on their flat (D44: shear across the layers). Keep the 5 spare pins with the base. A broken pin stub comes out with an M3 screw turned into it. Then slide the REEL gear onto the coupler journal (plain face first) and screw the retaining washer to the journal end (2 x M3 x 6): the gear must spin free.

Parts: drum_coupler_hub, drum_coupler, drum_shear_pin, drum_gear_reel, drum_reel_washer.

### Step 12 (main_shaft 131). Main shaft: index contact

Battery unplugged. Press an M3 nut into the index block pocket, screw the index lever foot to the block (M3 x 12), thread an M3 x 6 into the lever boss, head toward the nut. Screw the block's flange to the motor plate front face with two M3 x 10 into the frame inserts, nub on the coupler cam (D49: SW_INDEX stays on the frame, never on the rail). Set the gap so the screw head touches the nut near the top of a bump (about 1.5 mm on a low part). Battery unplugged: nut to GND, screw to SW_INDEX.

Parts: drum_index_block, drum_index_lever.

### Step 13 (main_stack 140). Main shaft: stack

Slide the collar onto the main shaft from the spigot end and push the shaft about 42 mm into the coupler socket. The shaft is held endwise later by the launch sleeve and the magnet cap.

Parts: drum_shaft, drum_collar.

### Step 14 (main_stack 141). Main shaft: LAUNCH dog ring and magnet cap

Fit the LAUNCH dog ring on the launch sleeve flange (sleeve lugs in the ring slots, 3 x M3 x 10 from the sleeve side into the ring) before the launch sleeve goes on the spigot (launcher steps). After the launcher: glue the AS5600 magnet into the cap, push the cap on the spigot end (flat to flat) and fit the M3 x 6 onto the flat.

Parts: drum_dogring_launch, drum_magnet_cap.

### Step 15 (launcher 190). Launcher 1: print the parts

All PLA unless named. PPA-CF: big gear, drive gear, launch sleeve, big gear pin, SW_SEATED plunger and pushrod, and the aim gears (sun, idler, output, bevel, worm, crank). Use each part's print orientation from the print sheet: the carriage body lies on its side across the bed diagonal; the stop springs print upside down at 100 percent infill (all perimeters); rail pieces lie on their largest flat face. 0.2 mm layers, 4 walls on every loaded part. Hardware (launcher and aim train): M3 screws 2 x 6, 3 x 8, 14 x 10, 23 x 12, 5 x 16, 12 x 20, 4 x 30, plus 4 x 10 for the aim housing to the deck (Aim 3); 10 M3 nuts, 1 M3 nyloc; M8 x 25 bolt and M8 nut; 2 x 608; 1 x 693ZZ (lip roller); 4 M3 heat-set inserts (rail rear flange for the reel head, D49); 2 x 8 mm steel rod pins (cut to the length on the print sheet) for the spring anchors; sprag #1 (CSK20PP); the two springs.

### Step 16 (launcher 191). Launcher 2: join the rail

Lay the rear, mid and front rail pieces in a line on a flat table. Push each joint together and fit 2 x M3 x 20 with nuts through each flange pair (8 in total). Tighten evenly; file any step inside the track where the pieces meet so the carriage shoes slide over it.

Parts: launch_rail_rear, launch_rail_mid, launch_rail_front.

### Step 17 (launcher 192). Launcher 2b: splice straps and aim lug

Screw a splice strap on each wall top across the front joint (4 x M3 x 10 self tap each). Bolt the aim lug to the left rail wall now (2 x M3 x 12 self tap): the spring guards cover its screws later.

Parts: launch_rail_splice_l, launch_rail_splice_r, launch_aim_lug.

### Step 18 (launcher 193). Launcher 3: big gear

Hold the big gear between the two cheeks of the mid piece, the toothed sector towards the drum side (right) and the full ring of teeth towards the left, with the spacer washer between the gear and the left cheek. Push the big gear pin through cheek, gear, washer, cheek from the right and snap the C clip into its groove. The gear must spin freely with almost no end play.

Parts: launch_big_gear, launch_gear_pin, launch_gear_clip, launch_gear_washer.

### Step 19 (launcher 194). Launcher 4: drive gear, sprag #1 and the launch sleeve

First bolt cad-drum's LAUNCH dog ring to the sleeve flange: lugs in the ring slots, 3 x M3 x 8 from the flange side into the tapped ring (the heads are out of reach once the sleeve is in the cheeks). Slide sprag #1 into the drive gear so its outer keyway takes the rib inside the gear. Check the direction: look at the gear from its collar side (the side that goes to the right, towards the motor), hold the gear and turn the inner race with a finger. It must turn freely counter-clockwise (BACKWARD) and lock clockwise (FORWARD: clockwise seen from the motor side is the cock direction). If it is the other way round, push the sprag out, turn it over and slide it in other face first. Hold the drive gear between the cheeks (collar side right, then the spacer washer on its left), meshing the big gear's full ring of teeth. Turn the cheek bores so their slots point down, then push the launch sleeve in from the motor side, outboard end first, ribs lined up with the slots and the sprag keyway, until the long (15 mm) rib sits inside the sprag.

Parts: launch_drive_gear, launch_sprag, launch_sleeve, launch_drive_washer.

### Step 20 (launcher 195). Launcher 5: carriage

Turn the big gear so its toothless gap faces the rack. Slide the carriage BODY into the track from the FRONT end (the rear flange blocks the rear), rack teeth down, until its rear face reaches the stop position. Yoke: push the SW_SEATED pushrod into the block bore from the rear, collar first, then push the plunger's pin up through the block from below (pad under the left arm) and press the head cap on the pin top. Lower the yoke onto the carriage rear so its key plates drop into the slots in the body webs, and fit 1 x M3 x 10 per side through the yoke into the webs. Press a printed spring peg up into each inner (y 38) peg hole from below, flat to the rear, to the end of the hole.

Parts: launch_carriage, launch_carriage_yoke, launch_seat_plunger, launch_seat_pushrod, launch_seat_clip, launch_yoke_peg_l, launch_yoke_peg_r.

### Step 21. Fit the remaining launch parts

Parts without an explicit assembly step: launch_dart.

Parts: launch_dart.

### Step 22 (launcher 196). Launcher 5b: lip roller

Press a 693ZZ into the lip roller, put a printed thrust washer each side in its bore (hub to the bearing), drop roller and washers into the fork at the carriage front and fit the M3 x 16 and a nyloc nut; it must spin freely. The tether runs over it in tow and retrieval, and the dart's tail slides in over it through the funnel onto the cradle.

Parts: launch_muzzle_roller, launch_muzzle_roller_brg, launch_muzzle_roller_axle, launch_muzzle_washer_l, launch_muzzle_washer_r.

### Step 23 (launcher 197). Launcher 6: carriage stop frames

For each side, before any spring: slide the stop frame onto the rail from the outboard side (it does not drop straight down past the inboard preload ledge). 2 x M3 x 12 through the rear foot into the rail wall, and 2 x M3 x 16 down through the front foot into the rail front plate, driven through the access holes in the frame's front wall.

Parts: launch_stop_frame_l, launch_stop_frame_r.

### Step 24 (launcher 198). Launcher 6b: spring guards

Screw the spring guard plates to the rail walls and anchor posts (M3 self tap into the pilots, see the print sheet), four per side: a (rear, with the end wall), b1 and b2 (either side of the stop frame's rear leg) and c (front, 2 x M3 x 12 into the front plate edge). Slide each one in from the side, under the stop frame's outboard bar. Wall brackets: M3 x 12 self tap, head and hex key through the hole in the plate; b2 foot: M3 x 10 down. Never cock with a guard off.

Parts: launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c.

### Step 25 (launcher 199). Launcher 6c: stop springs

Put each stop spring on its ledges inside its frame, rear bar towards the yoke arm, and pull it onto the frame's front wall with the 2 x M3 x 20 draw screws (heads on the front wall), a turn each in turn, until it sits on the wall (6.6 mm of preload). The rear bar must rest on both ledges.

Parts: launch_stop_spring_l, launch_stop_spring_r.

### Step 26 (launcher 200). Launcher 7: contacts

Battery unplugged. SW_HOME (right lip) and SW_SEATED (left lip): drop an M3 nut into the slot in the rail wall with a stripped wire end under it. Put a ring lug (or a stripped wire end wrapped once) on an M3 x 6 under its head and screw it up into the lever's boss, head down, until the tip is flush with the boss top (the pad beside it must stay the highest point), then screw the lever root to the lip with one M3 x 10 (self tap). Set the gap with the carriage pushed back at least 10 mm (lever free): turn the screw down until the head just touches the nut (meter beeps), then back it off ONE full turn (0.5 mm gap; a dab of nail varnish on the thread holds it). Carriage on its stop: the right foot presses the pad about 0.95 mm and SW_HOME must read closed; 2 mm back it must read open (if not, a quarter turn at a time). SW_SEATED closes only with a dart on the sled (no springs yet: hold the carriage on its stop by hand to check). SW_GAP, SW_RAIL_LOW and the AS5600 carrier are fitted in the Sensors steps (cad-sensors).

Parts: launch_sw_home, launch_sw_home_screw, launch_sw_home_nut, launch_sw_seated, launch_sw_seated_screw, launch_sw_seated_nut.

### Step 27 (launcher 201). Launcher 8: time the big gear (fingers clear)

No springs fitted yet. Hold the carriage on its stop by hand and turn the drive gear FORWARD until the first tooth after the toothless gap just touches the rearmost rack tooth: this is HOME (SW_GAP is set to it in the Sensors steps). Turning on pulls the carriage back smoothly; at full cock the cut last tooth lets go (push the carriage home by hand). Hold the drive gear from the side and never put a hand in front of the carriage.

### Step 28 (aim 205). Aim 1: aim housing, worm, wheel, crank

Build the aim housing on the bench (it goes onto the base later, after the rail). The AS5600 carrier and SW_RAIL_LOW are cad-sensors parts (Sensors steps), screwed into the pilots on the housing. Push the countershaft in from the inboard side through the inner wall bushing, through the big bevel (D bore, slid on from the outboard end of the shaft) and into the outer wall bushing. Drop the M8 nut into the hex pocket on the inner face of the outer wall, fit the two 608 bearings into the crank (worm wheel) hub, hold the crank in place and push the M8 x 25 bolt in from outboard through the wall, the nut and both bearings; tighten so the head and nut clamp the wall. Lay the worm shaft with the small bevel in its saddles and screw the cap on (4 x M3 x 30 self tap). The housing screws to the deck later (Aim 3), from the top: 4 x M3 x 10 down through the counterbored holes in its foot flanges (two behind the spine, one in front of the outer wall, one on the inboard tab) into the deck's heat-set inserts. The SW_RAIL_LOW bracket on the foot extension has its own deck screw (Sensors 2).

Parts: launch_aim_housing, launch_aim_cap, launch_aim_worm, launch_aim_crank, launch_aim_bolt, launch_aim_brg_1, launch_aim_brg_2, launch_aim_output, launch_aim_bevel30, launch_aim_nut.

### Step 29 (aim 206). Aim 2: splitter

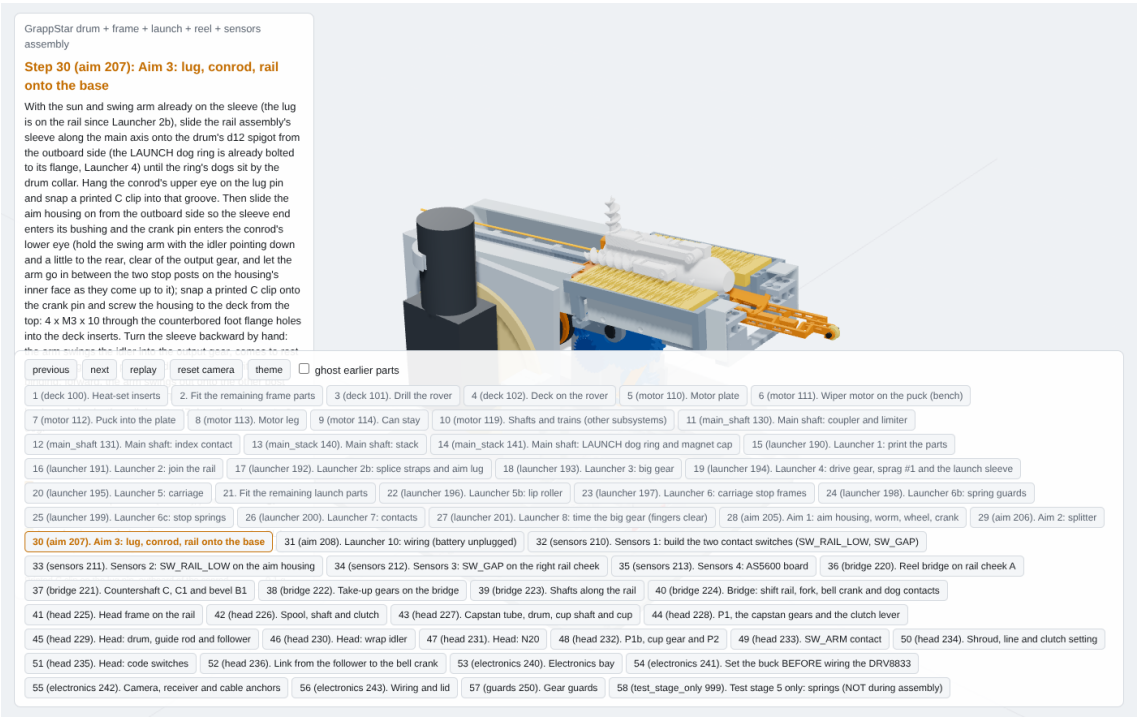
Slide the sun onto the sleeve's outer rib. On the bench: fit the idler on the swing arm pin and snap the printed C clip into the pin groove, and push the printed drag pad into the arm hub pocket with the shim behind its root block (a thicker shim gives more drag). Then slide the arm onto the sleeve next to the sun (there is no room to fit the idler once the arm is on).

Parts: launch_aim_sun, launch_aim_arm, launch_aim_idler, launch_aim_drag_pad, launch_aim_drag_shim, launch_aim_idler_clip.

## Step 30 (aim 207). Aim 3: lug, conrod, rail onto the base

With the sun and swing arm already on the sleeve (the lug is on the rail since Launcher 2b), slide the rail assembly's sleeve along the main axis onto the drum's d12 spigot from the outboard side (the LAUNCH dog ring is already bolted to its flange, Launcher 4) until the ring's dogs sit by the drum collar. Hang the conrod's upper eye on the lug pin and snap a printed C clip into that groove. Then slide the aim housing on from the outboard side so the sleeve end enters its bushing and the crank pin enters the conrod's lower eye (hold the swing arm with the idler pointing down and a little to the rear, clear of the output gear, and let the arm go in between the two stop posts on the housing's inner face as they come up to it); snap a printed C clip onto the crank pin and screw the housing to the deck from the top: 4 x M3 x 10 through the counterbored foot flange holes into the deck inserts. Turn the sleeve backward by hand: the arm swings the idler into the output gear, comes to rest on its engaged stop post and drives the gear without binding; forward, the arm swings out onto the other post and the output gear stays still. Then turn the wiper backward slowly: the rail rises to 47 deg, then lowers to 6 deg.

Parts: launch_aim_conrod, launch_aim_crankpin_clip, launch_aim_lugpin_clip.




## Step 31 (aim 208). Launcher 10: wiring (battery unplugged)

Battery unplugged. Run the SW_HOME, SW_SEATED and SW_GAP wires along the rail to the pivot and leave a 150 mm loop at the main axis before they go to the frame. No wire goes onto the carriage. Each contact: one wire to its ESP32 pin (SW_HOME 21, SW_GAP 39, SW_RAIL_LOW 40, SW_SEATED 41, as params.SWITCHES), the other to GND.

## Step 32 (sensors 210). Sensors 1: build the two contact switches (SW_RAIL_LOW, SW_GAP)

Battery unplugged. For each switch: press an M3 nut into the hex trap on the base pillar (a drop of glue), solder a wire to the nut through the side slot and lay it in the groove. Screw the lever onto the pad with 2 x M3x10 self tapping screws. Screw an M3x10 through the lever boss from the top, head down, with a ring lug and the second wire under a jam nut on top. Turn the screw until the head touches the nut, then back it off 1.9 turns (gap about 0.9 mm, 0.5 mm per turn) and lock the jam nut.

Parts: sensors_sw_rail_low, sensors_sw_rail_low_lever, sensors_sw_rail_low_screw, sensors_sw_rail_low_jam, sensors_sw_rail_low_nut.

### Step 33 (sensors 211). Sensors 2: SW_RAIL_LOW on the aim housing

Screw the switch base to its bracket (4 x M3x8 self tapping), lever pointing to the crank. Set the bracket foot on the housing foot extension in front of the crank. Screw it down with an M3x10 in the hole nearest the crank (self taps into the housing) and an M3x16 in the counterbored hole at the front end of the foot: down through the foot and the foot extension into the deck insert under it (the bracket's own deck screw). Turn the crank by hand (wiper backwards) to the lowest rail position: the cam lobe presses the lever tip and the meter shows the switch closed. Turn the crank about 20 deg either side: the switch must open.

Parts: sensors_sw_rail_low_bracket.

### Step 34 (sensors 212). Sensors 3: SW_GAP on the right rail cheek

Build SW_GAP as in Sensors 1 (its base has its own arm). Hold the arm tab on the outer face of the right rail cheek, below the big gear, and screw it on with 2 x M3x12 (heads in the counterbores). With the big gear at rest (toothless gap at the rack) the cam bump presses the lever: switch closed. Turn the gear 20 deg either way: switch open.

Parts: sensors_sw_gap, sensors_sw_gap_lever, sensors_sw_gap_screw, sensors_sw_gap_jam, sensors_sw_gap_nut.

### Step 35 (sensors 213). Sensors 4: AS5600 board

Battery unplugged. Do this after the magnet cap is on the main shaft end (drum steps): the board sits 2 mm in front of it. Battery unplugged. Plug the STEMMA QT cable into the board's connector on the +x end (the end away from the housing spine). Push the board onto the 4 pegs of the carrier, chip facing out. Drop an M3 nut into each hex pocket in the carrier back, lay the strips over the board edges (pegs through their holes) and screw each from the strip front with an M3x12 into its nut. Screw the carrier column to the aim housing spine with 2 x M3x16. Check by eye: the chip sits on the shaft axis, about 2.0 mm from the magnet face, and the cap turns free.

Parts: sensors_as5600, sensors_as5600_carrier, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_as5600_nut_top, sensors_as5600_nut_bot, sensors_as5600_screw_top, sensors_as5600_screw_bot, sensors_as5600_col_screws.

### Step 36 (bridge 220). Reel bridge on rail cheek A

Press the 8 mm c1' stub into its hole in the bridge plate. Press the CSK20PP into the sprag hub (key slot on the printed rib; seen from the outboard side the inner ring must turn freely counter-clockwise and lock clockwise) and screw the hub to the plate's outboard face with four M3 x 16. Hold the bridge inboard of the main shaft stack, below the REEL gear, and bolt its box arm to the inner face of rail cheek A with two M3 x 16 (heads reached through the two holes in the arm).

Parts: reel_bridge, reel_stub, reel_sprag_hub, reel_sprag.

### Step 37 (bridge 221). Countershaft C, C1 and bevel B1

Slide countershaft C into the sprag from the outboard side, key rib in the keyway, until its collar meets the plate. Slide C1 (49 teeth) onto the D-flat so it meshes the REEL gear, then the 24-tooth bevel, teeth outboard. Turn the REEL gear by hand: C1 turns one way only (the sprag holds the other).

Parts: reel_cs_c, reel_c1, reel_b1g.

### Step 38 (bridge 222). Take-up gears on the bridge

Slide c1' (46 teeth) onto the stub, hub outboard, meshing the bus gear. Clip the swing arm's split ring over c1's hub and put the 30-tooth swing idler on the arm pin. Push countershaft T into its bushing and slide T1 (20 teeth) and the B2 bevel onto its flat. Turn the wiper forward by hand: the idler swings into T1; backward it swings out.

Parts: reel_c1p, reel_swing_arm, reel_gi, reel_cs_t, reel_t1, reel_b2g.

### Step 39 (bridge 223). Shafts along the rail

Slide the 12-tooth B1 pinion onto the front flat of u-shaft C and feed the shaft rearward through the bridge block until the pinion meshes the B1 bevel. Slide the B2 pinion onto the front flat of u-shaft T and feed it rearward through the bridge pillar until the pinion meshes B2. Both shafts reach back to the reel head.

Parts: reel_b1p, reel_uc, reel_b2p, reel_ut.

### Step 40 (bridge 224). Bridge: shift rail, fork, bell crank and dog contacts

Battery unplugged. Press the shift rail into the bridge plate boss (flat facing the rail). Press an M3 nut into the hex hole of the fork's tab. Thread an M3 x 8 into each dog post boss, heads toward each other. Slide the fork sleeve onto the rail with the yoke under the collar and drop the yoke pads into the collar groove, then slide the dog post onto the rail end and tighten its clamp screw (M3 x 8, tapped) on the rail flat. Put the bell crank on the shelf with its pin up in the fork lug's slot and fit the M3 x 16 pivot screw (tapped into the shelf, the crank must swing). Battery unplugged: nut wire to GND, REEL side screw to SW_DOG_R, LAUNCH side screw to SW_DOG_L; the wires run in the flex loop to the frame.

Parts: drum_shift_rail, drum_fork, drum_dog_post, drum_crank.

### Step 41 (head 225). Head frame on the rail

Screw the front plate to the spine (three M3 x 12 through the spine into the plate). Slide the front plate's bushing over the rear end of u-shaft T and bolt the plate to the rail's rear flange with four M3 x 16 into the flange inserts. Screw the mid plate to the front plate (M3 x 12).

Parts: reel_head_front, reel_head_spine, reel_head_mid.

### Step 42 (head 226). Spool, shaft and clutch

Press a 608 into each end of the spool. Push the line end through the small hole near the inboard flange, knot it inside (figure eight) and wind 20 m on (line over the top, leaving toward the rail). Slide the clutch disk onto the outboard flat of the spool shaft (friction ring inboard), then from the other end the spool until its outboard face meets the disk, then the thrust ring, boss toward the spool. Feed the inboard end through the mid plate bushing and slide the B3 bevel onto its flat. Slide the B3 bevel for u-shaft T onto the shaft's rear flat; the two bevels mesh.

Parts: reel_spool, reel_brg_1, reel_brg_2, reel_s, reel_d, reel_t, reel_b3a, reel_b3b.

### Step 43 (head 227). Capstan tube, drum, cup shaft and cup

Slide the capstan drum onto the front end of the capstan tube, key slot on the rib. Push the cup shaft through the tube from the front (neck end in front), slide the cup onto it (flat in the bore onto the flat, eyelet lug at the bottom, in line with the spool) and push Harry's small C-clip onto the neck.

Parts: reel_c, reel_drum, reel_k, reel_cup, reel_cup_clip.

### Step 44 (head 228). P1, the capstan gears and the clutch lever

Press the G_b axle (8 mm rod, 26 long) into P1. Slide P1 over the capstan tube, the rear end of u-shaft C and the spool shaft end (its leg), and screw it to the spine (two M3 x 12) and to the mid plate. Put G_c (23 teeth) on the tube flat, G_a (26) on u-shaft C's flat and G_b (32) on its axle. Hang the clutch lever on the P1 post with the M3 x 16 pivot screw (it must swing), its upper arm under the drum's brake cam, the FIRM screw toward the thrust ring. Screw the LIGHT stop screw into P1 under the upper arm.

Parts: reel_head_p1, reel_gb_axle, reel_gc, reel_ga, reel_gb, reel_lever, reel_lever_pin, reel_firm_screw, reel_light_screw.

### Step 45 (head 229). Head: drum, guide rod and follower

Press the drum axle and the follower guide rod into the reel head front plate (flats as on the print sheet). Slide drum A onto the axle, gear first, then the follower onto the rod with its pin in drum A's groove, then drum B (hex socket over the spigot; it fits one way only). Press bracket B onto the axle and rod ends with the wrap idler shaft in its journal and screw the bracket's foot to the spine (2 x M3 x 12 from outside the spine).

Parts: drum_axle, drum_rod, drum_drum_a, drum_follower, drum_drum_b, drum_bracket_b.

### Step 46 (head 230). Head: wrap idler

Push the 49-tooth output gear onto the idler shaft's flat (D bore). Feed the shaft into bracket B from the front so the small pinion sits at the drum's segment teeth and the lock cam over the lock ring; its rear journal goes into P2 when P2 goes on. Turn the drum to LAUNCH: the flat-sided lock cam must sit in the drum lock ring and the idler can no longer turn by hand.

Parts: drum_idler, drum_idler_gear.

### Step 47 (head 231). Head: N20

Battery unplugged. Press the pinion onto the N20 output shaft (flat to flat) leaving 1.5 mm of shaft below it (that end runs in the front plate slot). Slide the N20 into the cradle from the open end, fit the end plate (2 x M3 x 8). Press an M3 nut into the pocket in the front plate's rear face, wire along its groove. Screw the cradle to the front plate: pivot screw M3 x 12 through the round hole, swing screw M3 x 12 through the arc slot at its full-mesh end, both tapped. Turn the SW_N20 screw (M3 x 8, tapped in the cradle plate) in until its tip just touches the nut. Strap the N20 body to the cradle with a cable tie (it rides the rail and sees every shot). Battery unplugged: nut to GND, screw to SW_N20.

Parts: drum_n20_pinion, drum_n20, drum_n20_cradle, drum_n20_strap.

### Step 48 (head 232). P1b, cup gear and P2

Fit P1b over the rear end of the capstan tube and screw it to the two P1 posts (M3 x 12). Slide the cup gear G_k (28 teeth, raised track forward) onto the cup shaft flat so it meshes the drum's wrap idler output with the cup at home (eyelet at the bottom). Screw P2 to the spine over the cup shaft and wrap idler ends (two M3 x 12). The cup must turn freely both ways.

Parts: reel_head_p1b, reel_gk, reel_head_p2.

### Step 49 (head 233). SW_ARM contact

Battery unplugged. Build the printed contact (see the contact switch page): nut pressed in the base trap, lever screwed onto the pad with 2 x M3 x 10 self tapping screws, M3 x 10 contact screw with its jam nut in the lever. Screw the base to P1b's rear face with four M3 x 10. SW_ARM is closed while the cup is away from home and opens at home: the lever rides the raised track on the front of the cup gear and drops into its notch at home. Cup at home: turn the contact screw until its head just touches the nut, back it off 1.9 turns (gap about 0.9 mm) and lock the jam nut. Check with the meter: about 15 degrees of cup turn either way closes it; at home it must be open.

Parts: reel_sw_base, reel_sw_lever, reel_sw_hw.

### Step 50 (head 234). Shroud, line and clutch setting

Battery unplugged. Put the shroud round the spool, open side at the top toward the rail, and screw its tab to the P1 leg (M3 x 8). Lead the line from the top of the spool straight forward under the capstan drum and through the cup eyelet from behind, then forward through the pusher hole to the dart's tail cap. Wind 20 m through the eyelet once on the bench before the first shot (D49: the pack mounds in the middle). Set LIGHT with the drum at LAUNCH: turn the LIGHT stop screw until a 25 g weight hung on the line end just pulls line off slowly (0.25 N). Set FIRM with the drum at NEUTRAL: turn the FIRM screw until the line slips at 5.0 N on the luggage scale, then lock its nut.

Parts: reel_shroud, launch_line_p, launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8, launch_line_a9, launch_line_a10, launch_line_a11, launch_line_a12, launch_line_a13, launch_line_a14, launch_line_a15, launch_line_a16, launch_line_a17, launch_line_a18, launch_line_a19, launch_line_a20.

### Step 51 (head 235). Head: code switches

Battery unplugged. Press an M3 nut into each of the 3 pockets of the switch block and screw its foot to the spine (2 x M3 x 12 from outside the spine). Screw each lever foot to the block (M3 x 12), nub on its band. Thread an M3 x 6 into each lever boss, head toward the nut, so the head touches the nut near the top of a bump (gap about 1.8 mm with the nub low). Battery unplugged: nuts to GND, screws to SW_DRUM_A (nearest the drum gear), SW_DRUM_B, SW_DRUM_C.

Parts: drum_switch_block, drum_sw_lever_0, drum_sw_lever_1, drum_sw_lever_2.

### Step 52 (head 236). Link from the follower to the bell crank

Cut the 8 mm steel rod to the length on the print sheet, deburr the ends and press a printed eye onto each end, both eye holes parallel. Pass the rod through the window in the head front plate, screw the top eye to the follower lug and the bottom eye to the bell crank's input arm (M3 x 16 and nyloc nut each, snug: the eyes must swing). Turn the drum by hand from LAUNCH to REEL: the follower slides 14 mm, the crank turns and the collar moves from the LAUNCH dogs to the REEL dogs.

Parts: drum_link, drum_link_eye_1, drum_link_eye_2.

### Step 53 (electronics 240). Electronics bay

Battery unplugged. Bay on the deck, 4 x M3 x 10 through the tabs. Boards in: BTS7960 on its 4 standoffs (M3 x 6), heatsink down; ESP32 DevKitC pins down into the cradle, USB end into the wall slot, clamp with 2 x M3 x 8; DRV8833, MP1584 and HX711 into their pockets with a cable tie through the floor slots or a drop of hot glue. Fuse holder in its cradle with a cable tie.

Parts: frame_ebay_lid, frame_ebay_esp_clamp, frame_pcb_esp32, frame_pcb_bts7960, frame_pcb_mp1584, frame_pcb_drv8833, frame_pcb_hx711, frame_fuse_20a.
Also handle (fitted earlier): frame_ebay.

### Step 54 (electronics 241). Set the buck BEFORE wiring the DRV8833

Battery unplugged. Wire only the buck input to the fuse output and ground. Plug the battery in, meter on the buck output, turn the trimpot until it reads 9.0 V. Unplug the battery. Only then wire buck output to DRV8833 VIN (the DRV8833 dies above 10.8 V).

Also handle (fitted earlier): frame_pcb_mp1584, frame_pcb_drv8833.

### Step 55 (electronics 242). Camera, receiver and cable anchors

Battery unplugged. Camera column on the deck, 4 x M3 x 10 into the inserts. Clip the Kiyo X onto the 16 mm edge plate like onto a monitor (or use a 1/4-20 tripod screw up through the plate hole); it looks forward along the launch line. RC832: stand it antenna-end up in its holder slot, leads out of the low notch, holder 2 x M3 x 10. Cable anchors 1 x M3 x 12 each; tie the camera USB, the receiver AV lead and the capture dongle lead to them so no plug carries the cable weight.

Parts: frame_camera_mount, frame_rc832_holder, frame_rc832, frame_kiyo_x, frame_cable_anchor_1, frame_cable_anchor_2.

### Step 56 (electronics 243). Wiring and lid

Battery unplugged. Follow firmware/base/WIRING.md. Every wire leaving the bay goes through its wall exit and is cable-tied to the anchor beside it (strain relief). 14 AWG for battery, fuse, BTS7960 and motor. Lid on, 4 x M3 x 6. USB cable from the ESP32 native USB port to the Jetson.

Also handle (fitted earlier): frame_ebay_lid.

### Step 57 (guards 250). Gear guards

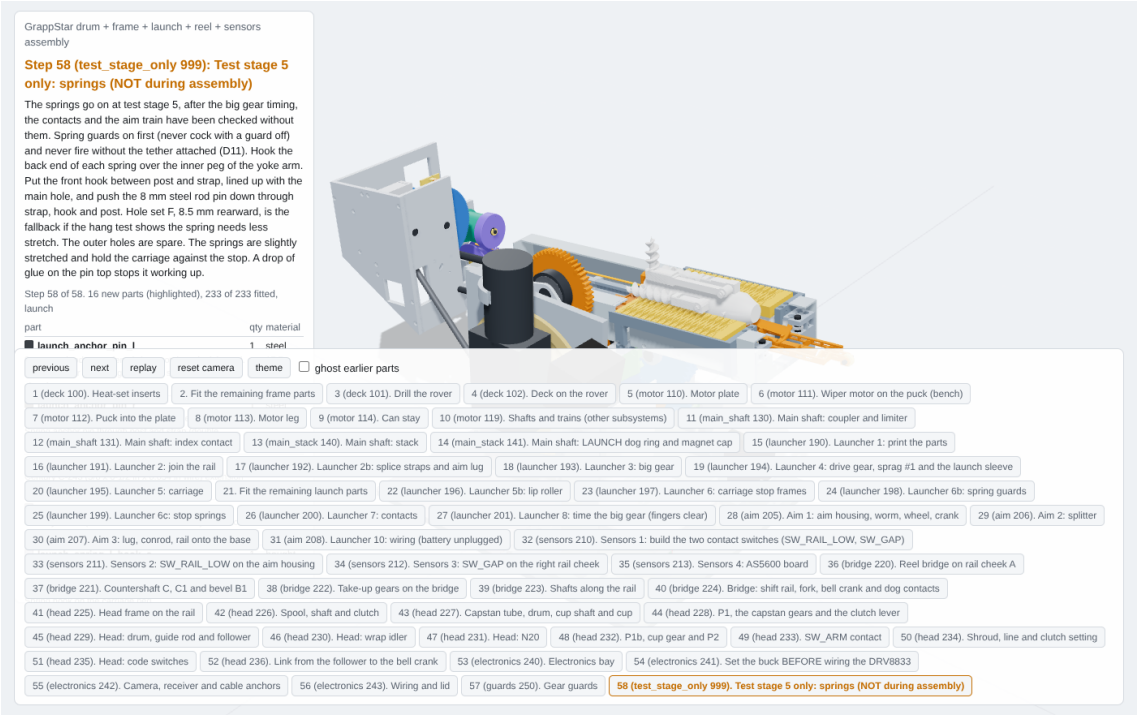
After every train has been turned by hand: the hood goes over the gears in front of the motor plate, M3 x 10 through each lug into the plate inserts; the front guard stands on the deck between the drum gears and the bay, M3 x 10 into the deck inserts. Turn the wiper by hand once: nothing rubs.

Parts: frame_guard_hood, frame_guard_front.

### Step 58 (test_stage_only 999). Test stage 5 only: springs (NOT during assembly)

The springs go on at test stage 5, after the big gear timing, the contacts and the aim train have been checked without them. Spring guards on first (never cock with a guard off) and never fire without the tether attached (D11). Hook the back end of each spring over the inner peg of the yoke arm. Put the front hook between post and strap, lined up with the main hole, and push the 8 mm steel rod pin down through strap, hook and post. Hole set F, 8.5 mm rearward, is the fallback if the hang test shows the spring needs less stretch. The outer holes are spare. The springs are slightly stretched and hold the carriage against the stop. A drop of glue on the pin top stops it working up.

Parts: launch_anchor_pin_l, launch_anchor_pin_r, launch_spring_l_hook_a, launch_spring_r_hook_a, launch_spring_l_hook_c, launch_spring_r_hook_c, launch_spring_l_0, launch_spring_l_1, launch_spring_l_2, launch_spring_l_3, launch_spring_l_4, launch_spring_r_0, launch_spring_r_1, launch_spring_r_2, launch_spring_r_3, launch_spring_r_4.




## 6. Audit summary

Result: NOT CLEAN

233 parts checked over 46676 states.

Collision drum_coupler with drum_index_lever: 24.517 mm3 in 67 states.

Collision launch_aim_sun with launch_aim_idler: 55.565 mm3 in 12 states.

Allowed contacts: 274.